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MIL-M-28787/85A(EC)
4 June 1982
SUPERSEDING
MIL-M-28787/85(EC)
25 March 1977

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE BDN

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.32 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/85-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table 1.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current and pullup resistor current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

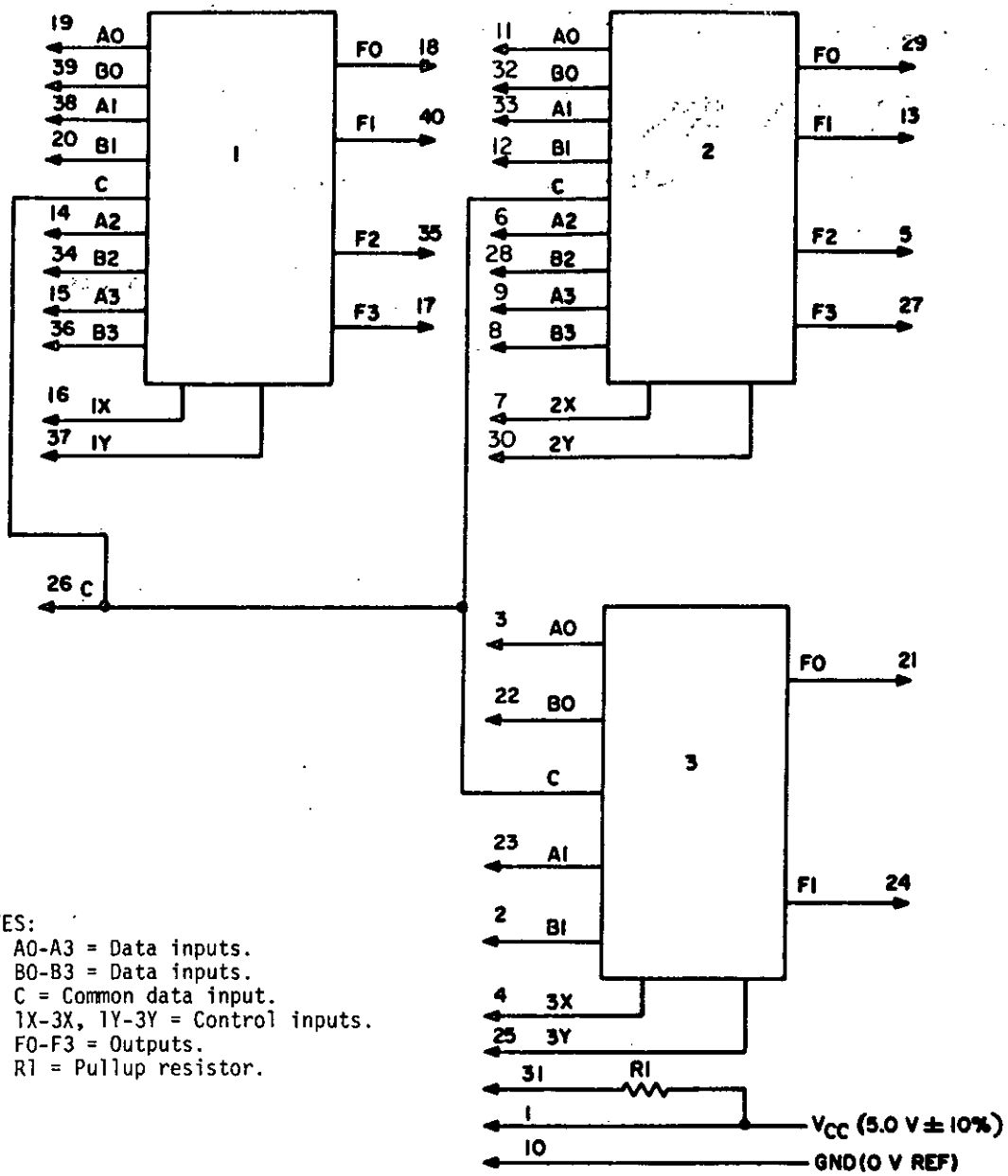
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. A0-A3 = Data inputs.
2. B0-B3 = Data inputs.
3. C = Common data input.
4. 1X-3X, 1Y-3Y = Control inputs.
5. F0-F3 = Outputs.
6. R1 = Pullup resistor.

FIGURE 1. BDN module function diagram, multiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B10	V_{OUT} (low-loaded)	IV	S	0.01	0.6	0.01	0.6	0.01	0.6	0.01	0.6	V
C1	Input breakdown current (C)	V	S	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	mA
C2-C13	Input breakdown current (A, 3X, and 3Y)	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C14-C23	Input breakdown current (B)	V	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	mA
C24-C27	Input breakdown current (1X, 1Y, 2X, and 2Y)	V	S	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	mA
D1-D10	Output short circuit current	VI	X	-10.0	-40.0	-10.0	-40.0	-10.0	-40.0	-10.0	-40.0	mA
D11	Pullup resistor current	VI	X	-1.2	-1.5	-1.2	-1.5	-1.2	-1.5	-1.2	-1.5	mA
E1-E32	Functional	VII	X	← See table VII for limits →								
F1-F10	Static output voltage, low level	VIII	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
G1-G10	Static output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H13	Input current, high level (A, C, 3X, and 3Y)	X	X	0.0	20.0	0.0	20.0	0.0	20.0	0.0	20.0	μA
H14-H23	Input current, high level (B)	X	X	0.0	30.0	0.0	30.0	0.0	30.0	0.0	30.0	μA
H24-H27	Input current, high level (1X, 1Y, 2X, and 2Y)	X	X	0.0	40.0	0.0	40.0	0.0	40.0	0.0	40.0	μA
J1-J12	Input current, low level (A, 3X, and 3Y)	XI	X	-0.10	-0.43	-0.10	-0.43	-0.10	-0.43	-0.10	-0.43	mA
J13	Input current, low level (C)	XI	X	-0.10	-0.40	-0.10	-0.40	-0.10	-0.40	-0.10	-0.40	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J14-J23	Input current, low level (B)	XI	X	-0.10	-0.63	-0.10	-0.63	-0.10	-0.63	-0.10	-0.63	mA
J24-J27	Input current, low level (1X, 1Y, 2X, and 2Y)	XI	X	-0.20	-0.86	-0.20	-0.86	-0.20	-0.86	-0.20	-0.86	mA
K1	Power supply current (inputs at high level)	XII	S	20	64	20	64	20	64	20	64	mA
K2	Power supply current (inputs at low level)	XII	S	18	60	18	60	18	60	18	60	mA
L1-L10	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	2	20	2	20	2	25	2	25	ns
L11-L20	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	2	40	2	40	2	45	2	45	ns
M1-M10	Transition time (t_{THL}), C_L = 50 pF	XIV	S	2	15	2	15	2	20	2	20	ns
M11-M20	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	2	30	2	30	2	35	2	35	ns
N1-N10	Propagation delay time (t_{PHL}), A to F, noninverting, C_L = 50 pF	XV	S	10	70	10	70	10	75	10	75	ns
N11-N20	Propagation delay time (t_{PHL}), B to F, noninverting, C_L = 50 pF	XV	S	10	70	10	70	10	75	10	75	ns
N21-N30	Propagation delay time (t_{PHL}), B to F, inverting, C_L = 50 pF	XV	S	20	110	20	110	20	120	20	120	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N31-N40	Propagation delay time (t_{PHL}), X to F, noninverting, C_L = 50 pF	XV	S	20	70	20	70	20	75	20	75	ns
N41-N50	Propagation delay time (t_{PHL}), X to F, inverting, C_L = 50 pF	XV	S	15	70	15	70	15	75	15	75	ns
N51-N60	Propagation delay time (t_{PHL}), Y to F, noninverting, C_L = 50 pF	XV	S	20	70	20	70	20	75	20	75	ns
N61-N70	Propagation delay time (t_{PHL}), Y to F, inverting, C_L = 50 pF	XV	S	15	70	15	70	15	75	15	75	ns
N71-N80	Propagation delay time (t_{PHL}), C to F, inverting, C_L = 50 pF	XV	S	20	115	20	115	20	120	20	120	ns
P1-P10	Propagation delay time (t_{PHL}), A to F, noninverting, C_L = 150 pF	XV	Q	10	85	10	85	10	90	10	90	ns
P11-P20	Propagation delay time (t_{PHL}), B to F, noninverting, C_L = 150 pF	XV	Q	10	85	10	85	10	90	10	90	ns
P21-P30	Propagation delay time (t_{PHL}), B to F, inverting, C_L = 150 pF	XV	Q	20	125	20	125	20	135	20	135	ns
P31-P40	Propagation delay time (t_{PHL}), X F, noninverting, C_L = 150 pF	XV	Q	20	85	20	85	20	90	20	90	ns
P41-P50	Propagation delay time (t_{PHL}), X to F, inverting, C_L = 150 pF	XV	Q	15	85	15	85	15	90	15	90	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P51-P60	Propagation delay time (t_{PHL}), Y to F, noninverting, C_L = 150 pF	XV	Q	20	85	20	85	20	90	20	90	ns
P61-P70	Propagation delay time (t_{PHL}), Y to F, inverting, C_L = 150 pF	XV	Q	15	85	15	85	15	90	15	90	ns
P71-P80	Propagation delay time (t_{PHL}), C to F, inverting, C_L = 150 pF	XV	Q	20	125	20	125	20	135	20	135	ns
Q1-Q10	Propagation delay time (t_{PLH}), A to F, noninverting, C_L = 50 pF	XVI	S	10	75	10	75	10	80	10	80	ns
Q11-Q20	Propagation delay time (t_{PLH}), B to F, noninverting, C_L = 50 pF	XVI	S	10	75	10	75	10	80	10	80	ns
Q21-Q30	Propagation delay time (t_{PLH}), B to F, inverting, C_L = 50 pF	XVI	S	20	115	20	115	20	120	20	120	ns
Q31-Q40	Propagation delay time (t_{PLH}), X to F, noninverting, C_L = 50 pF	XVI	S	20	75	20	75	20	80	20	80	ns
Q41-Q50	Propagation delay time (t_{PLH}), X to F, inverting, C_L = 50 pF	XVI	S	15	70	15	70	15	75	15	75	ns
Q51-Q60	Propagation delay time (t_{PLH}), Y to F, noninverting, C_L = 50 pF	XVI	S	20	75	20	75	20	80	20	80	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q61-Q70	Propagation delay time (t_{PLH}), Y to F, inverting, C_L = 50 pF	XVI	S	15	70	15	70	15	75	15	75	ns
Q71-Q80	Propagation delay time (t_{PLH}), C to F, inverting, C_L = 50 pF	XVI	S	20	115	20	115	20	120	20	120	ns
R1-R10	Propagation delay time (t_{PLH}), A to F, noninverting, C_L = 150 pF	XVI	Q	10	90	10	90	10	95	10	95	ns
R11-R20	Propagation delay time (t_{PLH}), B to F, noninverting, C_L = 150 pF	XVI	Q	10	90	10	90	10	95	10	95	ns
R21-R30	Propagation delay time (t_{PLH}), B to F, inverting, C_L = 150 pF	XVI	Q	20	130	20	130	20	135	20	135	ns
R31-R40	Propagation delay time (t_{PLH}), X to F, noninverting, C_L = 150 pF	XVI	Q	20	90	20	90	20	95	20	95	ns
R41-R50	Propagation delay time (t_{PLH}), X to F, inverting, C_L = 150 pF	XVI	Q	15	85	15	85	15	90	15	90	ns
R51-R60	Propagation delay time (t_{PLH}), Y to F, noninverting, C_L = 150 pF	XVI	Q	20	90	20	90	20	95	20	95	ns
R61-R70	Propagation delay time (t_{PLH}), Y to F, inverting, C_L = 150 pF	XVI	Q	15	85	15	85	15	90	15	90	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R71-R80	Propagation delay time (t_{PLH}), C to F, inverting, C_L = 150 pF	XVI	Q	20	125	20	125	20	135	20	135	ns
S	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	mV
T1-T10	Output crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
U1-U10	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W12	Input capacitance (A, 3X, and 3Y)	XXI	Q	2	15	2	15	2	15	2	15	pF
W13-W27	Input capacitance (B, C, 1X, 1Y, 2X, and 2Y)	XXI	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/85-1 supersedes part number M28787/85. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
All inputs at high level	352 mW (outputs not loaded)
All inputs at low level	330 mW (outputs not loaded)
Supply voltage	5.0 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.7 V maximum
High level	2.0 V minimum, V_{CC} maximum

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Input current		
C input		
Low level	(V_{IL} = 0.0 V)	-0.40 mA maximum
	(V_{IL} = 0.3 V)	-0.36 mA maximum
High level	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	200 µA maximum
A, 3X, or 3Y inputs		
Low level	(V_{IL} = 0.0 V)	-0.43 mA maximum
	(V_{IL} = 0.3 V)	-0.40 mA maximum
High level	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
B inputs		
Low level	(V_{IL} = 0.0 V)	-0.63 mA maximum
	(V_{IL} = 0.3 V)	-0.58 mA maximum
High level	(V_{IH} = 2.4 V)	30 µA maximum
	(V_{IH} = 5.5 V)	1.1 mA maximum
1X, 1Y, 2X, or 2Y inputs		
Low level	(V_{IL} = 0.0 V)	-0.86 mA maximum
	(V_{IL} = 0.3 V)	-0.80 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
Output voltage		
Low level (I_{OL} = 4.0 mA)		0.3 V maximum
High level (I_{OH} = -0.4 mA)		2.4 V maximum
Output drive current maximum capability		
Low level (V_{OL} = 0.3 V)		4.0 mA
High level (V_{OH} = 2.4 V)		-0.4 mA
Propagation delay time (t_{PHL})		
A or B to F (noninverting)		
C_L = 50 pF		10 ns minimum, 75 ns maximum
C_L = 150 pF		10 ns minimum, 90 ns maximum
B or C to F (inverting)		
C_L = 50 pF		20 ns minimum, 120 ns maximum
C_L = 150 pF		20 ns minimum, 135 ns maximum
X or Y to F (noninverting)		
C_L = 50 pF		20 ns minimum, 75 ns maximum
C_L = 150 pF		20 ns minimum, 90 ns maximum
X or Y to F (inverting)		
C_L = 50 pF		15 ns minimum, 75 ns maximum
C_L = 150 pF		15 ns minimum, 90 ns maximum
Propagation delay time (t_{PLH})		
A or B to F (noninverting)		
C_L = 50 pF		10 ns minimum, 80 ns maximum
C_L = 150 pF		10 ns minimum, 95 ns maximum
B or C to F (inverting)		
C_L = 50 pF		20 ns minimum, 120 ns maximum
C_L = 150 pF		20 ns minimum, 135 ns maximum
X or Y to F (noninverting)		
C_L = 50 pF		20 ns minimum, 80 ns maximum
C_L = 150 pF		20 ns minimum, 95 ns maximum
X or Y to F (inverting)		
C_L = 50 pF		15 ns minimum, 75 ns maximum
C_L = 150 pF		15 ns minimum, 90 ns maximum

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6.5 Module function. This module contains two quad 2-input multiplexer circuits with common select lines and one dual 2-input multiplexer with common select lines. A third input, common to all multiplexers, can be selected with the common select lines.

6.5.1 Input to output relationship. Each multiplexer has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Control inputs		Data inputs			Output
X	Y	A	B	C	F
L	L	X	X	H	L
L	L	X	X	L	H
H	L	X	H	X	L
H	L	X	L	X	H
L	H	X	H	X	H
L	H	X	L	X	L
H	H	H	X	X	H
H	H	L	X	X	L

1/ H = High-level voltage.
2/ L = Low-level voltage.
3/ X = Don't care.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

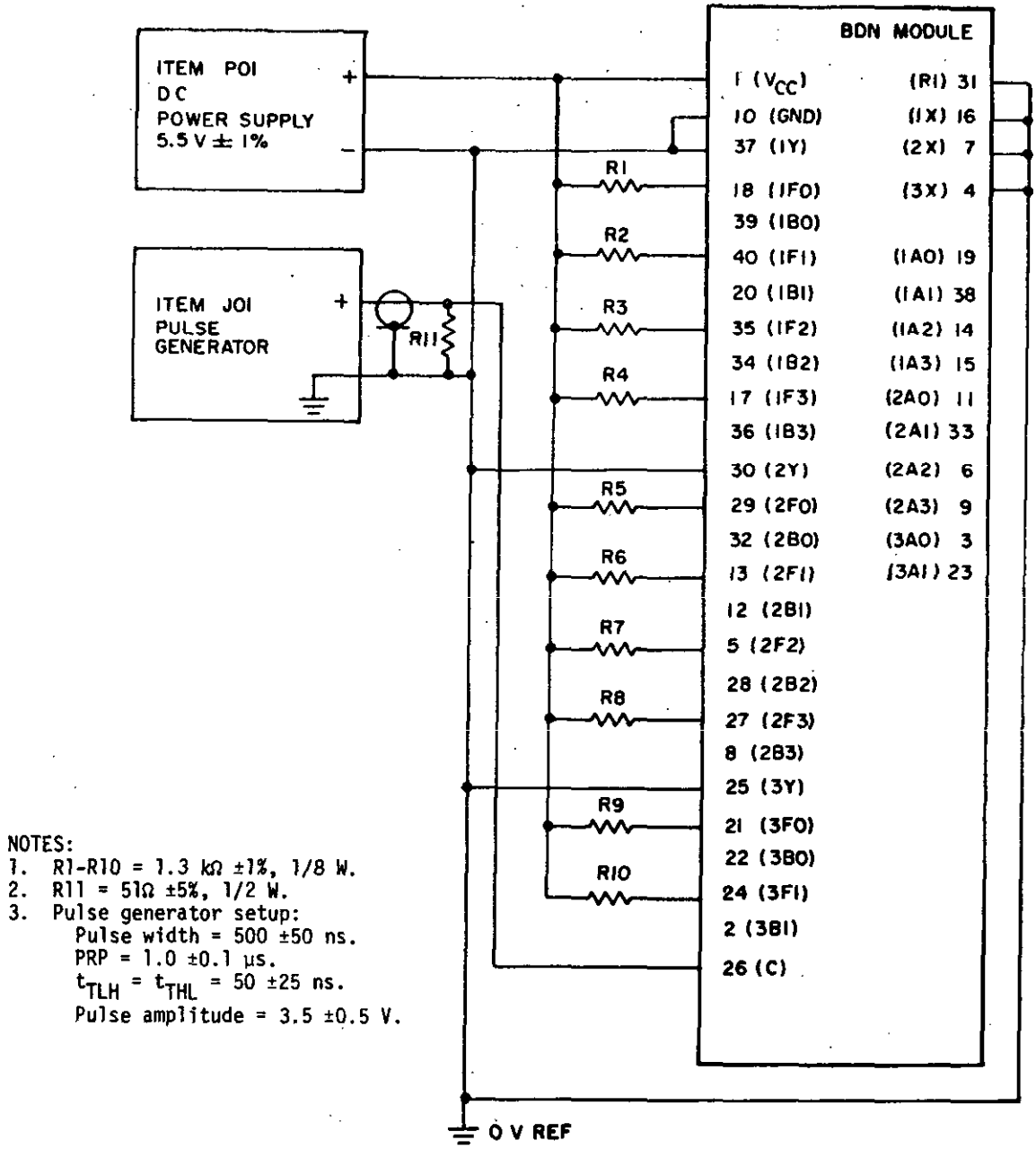
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N310)

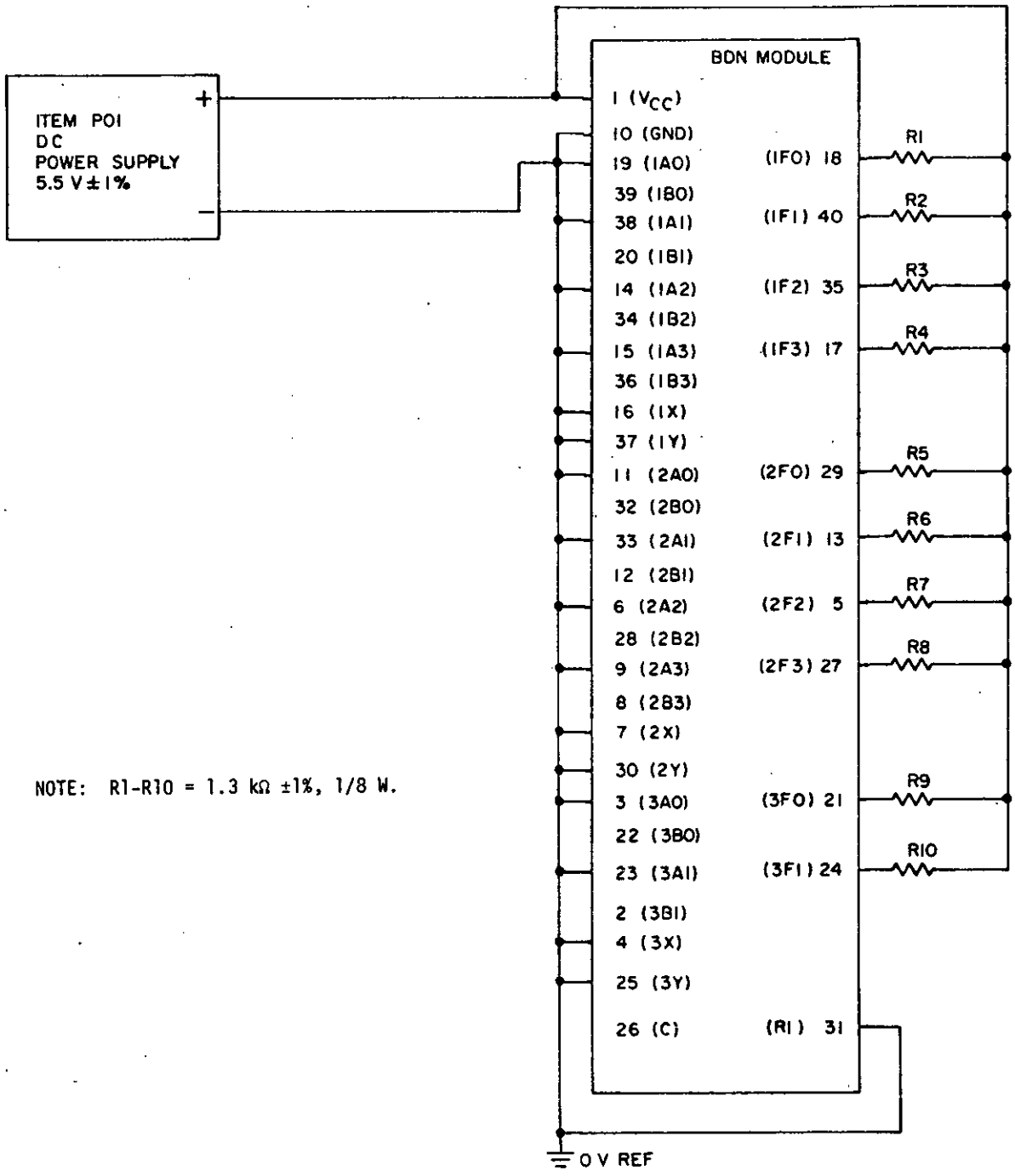
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NOTES:
1. R1-R10 = 1.3 kΩ ±1%, 1/8 W.
2. R11 = 51Ω ±5%, 1/2 W.
3. Pulse generator setup:
   Pulse width = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
   Pulse amplitude = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R10 = 1.3 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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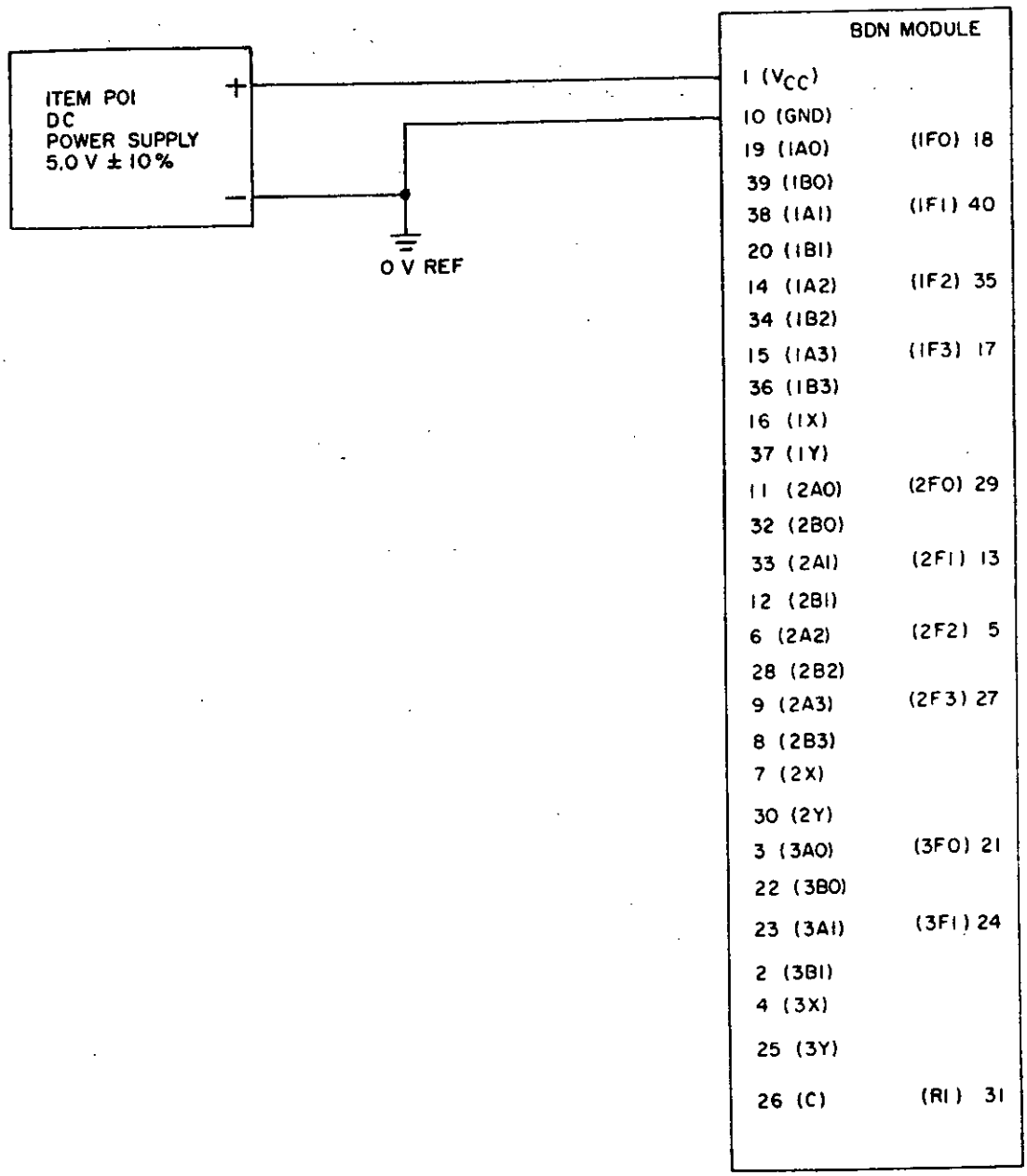


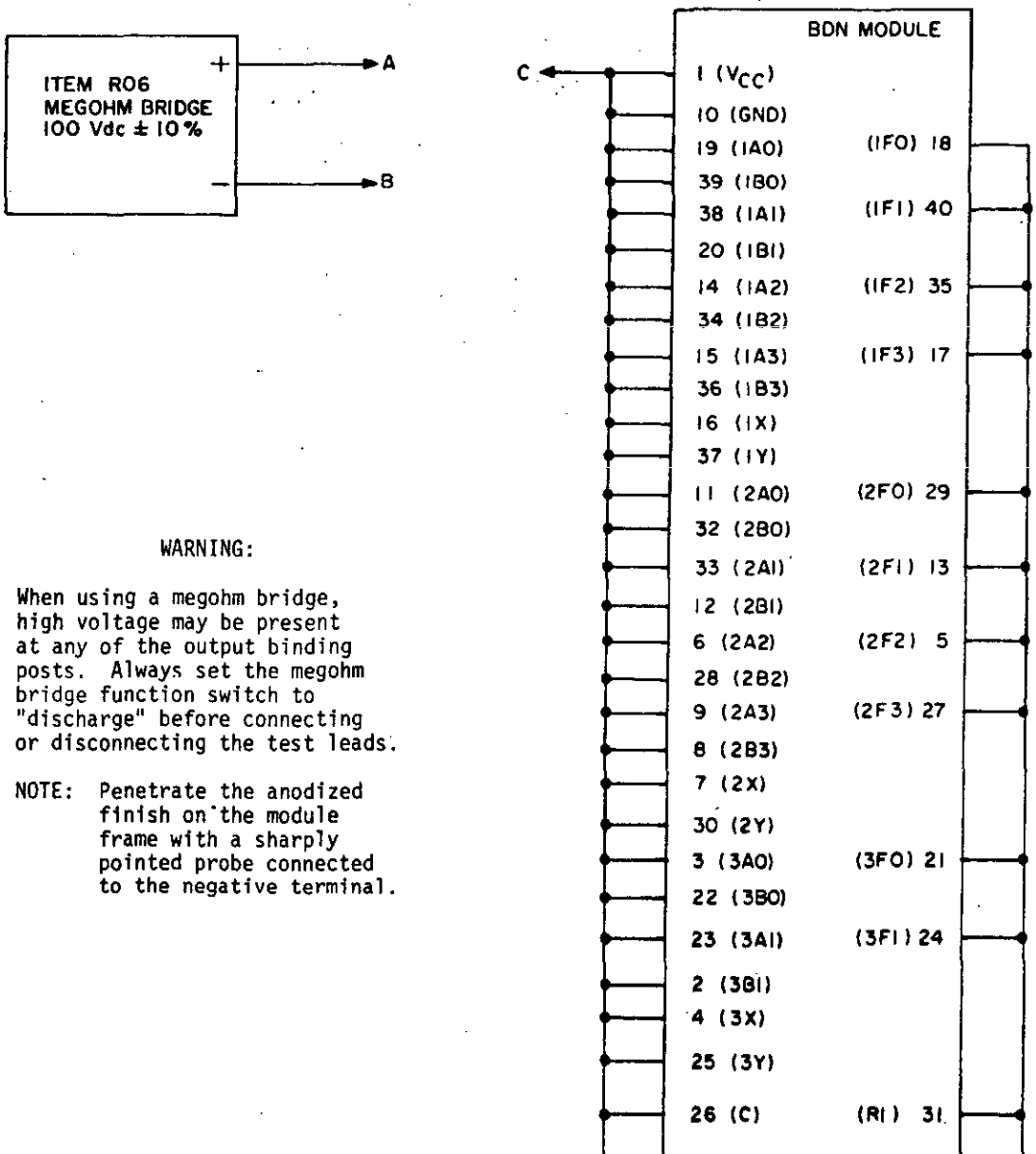
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

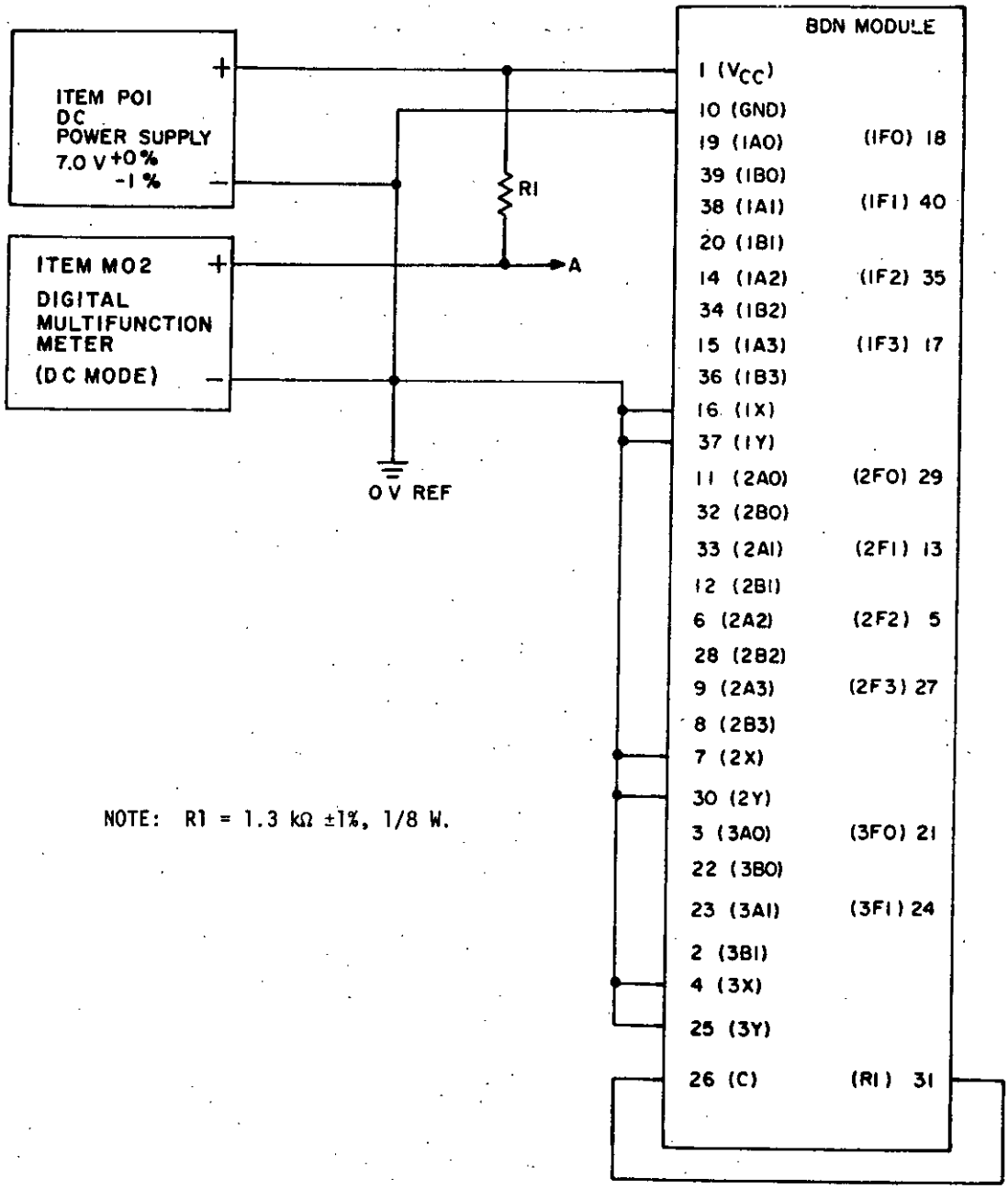
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1F0 V_{OUT} (low loaded)	18	0.01	0.6	V	6
B2	1F1	40	↓	↓	↓	↓
B3	1F2	35				
B4	1F3	17				
B5	2F0	29				
B6	2F1	13				
B7	2F2	5				
B8	2F3	27				
B9	3F0	21				
B10	3F1	24				

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NOTE: R1 = 1.3 kΩ ±1%, 1/8 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	C Input breakdown current	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,28,30,32,33, 34,36,37,38,39	26	0.0	0.2	mA	7
C2	1A0	2,3,4,6,7,8,9,11,12,14,15,16, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	19		1.0		
C3	1A1	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,39	38				
C4	1A2	2,3,4,6,7,8,9,11,12,15,16,19, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	14				
C5	1A3	2,3,4,6,7,8,9,11,12,14,16,19, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	15				
C6	2A0	2,3,4,6,7,8,9,12,14,15,16,19, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	11				
C7	2A1	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 34,36,37,38,39	33				
C8	2A2	2,3,4,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,38,39	6				
C9	2A3	2,3,4,6,7,8,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,38,39	9				

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TABLE V. Input breakdown current test - Continued.

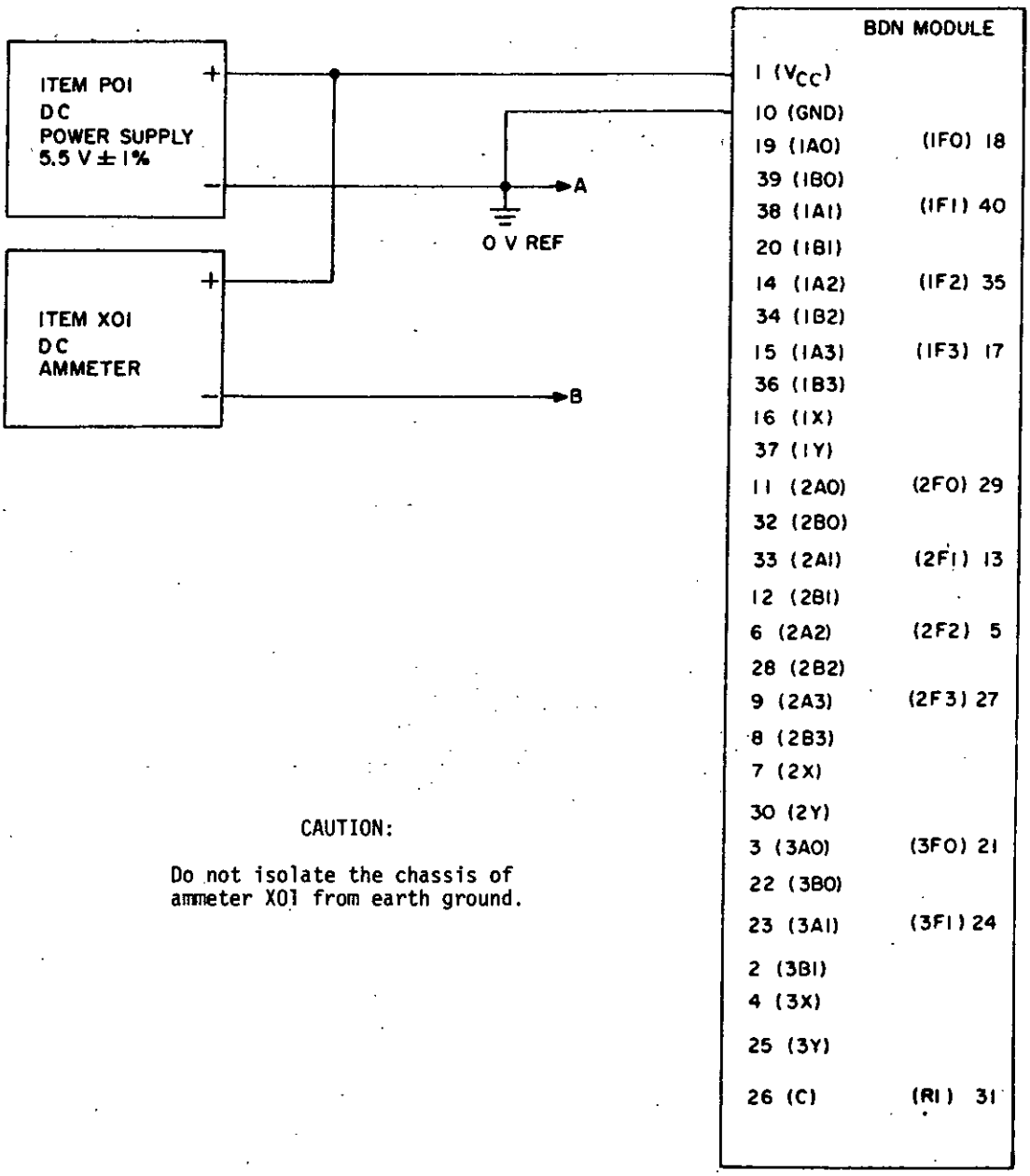
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C10	3A0 Input breakdown current	2,4,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,32,33,34,36,37,38,39	3	0.0	1.0	mA	7
C11	3A1	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,25,26,28,30,32,33,34,36,37,38,39	23				
C12	3X	2,3,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,32,33,34,36,37,38,39	4				
C13	3Y	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,23,26,28,30,32,33,34,36,37,38,39	25				
C14	1B0	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,32,33,34,36,37,38	39		1.1		
C15	1B1	2,3,4,6,7,8,9,11,12,14,15,16,19,22,23,25,26,28,30,32,33,34,36,37,38,39	20				
C16	1B2	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,32,33,36,37,38,39	34				
C17	1B3	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,32,33,34,37,38,39	36				
C18	2B0	2,3,4,6,7,8,9,11,12,14,15,16,19,20,22,23,25,26,28,30,33,34,36,37,38,39	32				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	2B1 Input breakdown current	2,3,4,6,7,8,9,11,14,15,16,19, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	12	0.0	1.1	mA	7
C20	2B2	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,30,32,33, 34,36,37,38,39	28				
C21	2B3	2,3,4,6,7,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,38,39	8				
C22	3B0	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,23,25,26,28,30,32,33, 34,36,37,38,39	22				
C23	3B1	3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,38,39	2				
C24	1X	2,3,4,6,7,8,9,11,12,14,15,19, 20,22,23,25,26,28,30,32,33, 34,36,37,38,39	16		2.0		
C25	1Y	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,38,39	37				
C26	2X	2,3,4,6,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32, 33,34,36,37,38,39	7				
C27	2Y	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,32,33, 34,36,37,38,39	30				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

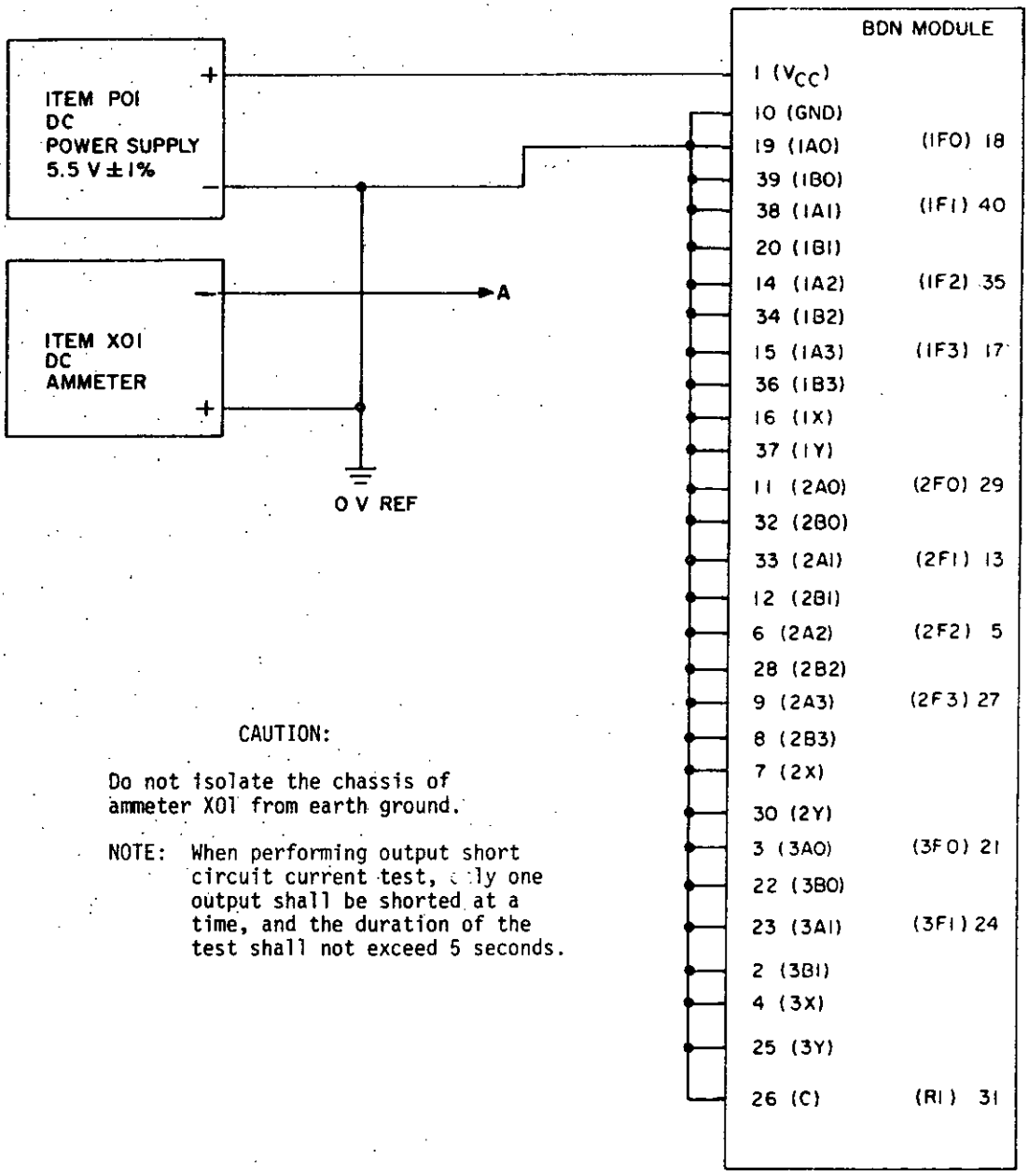
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current and pullup resistor current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1F0 Output short circuit current	18	-10.0	-40.0	mA	8
D2	1F1	40				
D3	1F2	35				
D4	1F3	17				
D5	2F0	29				
D6	2F1	13				
D7	2F2	5				
D8	2F3	27				
D9	3F0	21				
D10	3F1	24				
D11	R1 Pullup resistor current	31	-1.2	-1.5		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

FIGURE 8. Output short circuit current and pullup resistor current test circuit for table VI.

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TABLE VII. Functional test.

Test condition	Parameter	Connect			Output logic states 1/										Test figure
		A to pin	B to pin	C to pin	1F0	1F1	1F2	1F3	2F0	2F1	2F2	2F3	3F0	3F1	
E1	Functional	25,30,37,39	2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,26,28,32,33, 34,36,38	2/, 3/	H	L	L	L	L	L	L	L	L	L	9
E2		3,6,9,11,14,15,19, 20,23,25,30,33,37	2,4,7,8,12,16,22, 26,28,32,34,36,38, 39		L	H	L	L	L	L	L	L	L	L	
E3		25,26,30,34,37	2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,28,32,33,36, 38,39		L	L	H	L	L	L	L	L	L	L	
E4		3,6,9,11,14,19,23, 25,26,30,33,36,37, 38	2,4,7,8,12,15,16, 20,22,28,32,34,39		L	L	L	H	L	L	L	L	L	L	
E5		2,4,7,8,12,16,20, 22,28,34,36,39	3,6,9,11,14,15,19, 23,25,26,30,32,33, 37,38		L	L	L	L	H	L	L	L	L	L	
E6		2,3,4,6,7,8,9,11, 14,15,16,19,20,22, 23,28,32,34,36,38, 39	12,25,26,30,33,37		L	L	L	L	L	H	L	L	L	L	
E7		2,4,7,8,12,16,20, 22,26,32,34,36,39	3,6,9,11,14,15,19, 23,25,28,30,33,37, 38		L	L	L	L	L	L	H	L	L	L	
E8		2,3,4,6,7,11,12, 14,15,16,19,20,22, 23,26,28,32,33,34, 36,38,39	8,9,25,30,37		L	L	L	L	L	L	L	H	L	L	
E9		2,3,4,7,8,12,16, 20,25,28,30,32,34, 36,37,39	6,9,11,14,15,19, 22,23,26,33,38		L	L	L	L	L	L	L	L	H	L	

See footnotes at end of table.

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TABLE VII. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states 1/										Test figure
		A to pin	B to pin	C to pin	1F0	1F1	1F2	1F3	2F0	2F1	2F2	2F3	3F0	3F1	
E10	Functional	4,7,16,23,25,30,37	2,3,6,8,9,11,12, 14,15,19,20,22,26, 28,32,33,34,36,38, 39	2/, 3/	L	L	L	L	L	L	L	L	L	H	9
E11		2,8,12,22,25,28, 30,32,34,36,37,38	3,4,6,7,9,11,14, 15,16,19,20,23,26, 33,39		L	L	H	H	H	H	H	H	H	H	
E12		2,8,12,15,20,22, 25,26,28,30,32,37, 39	3,4,6,7,9,11,14, 16,19,23,33,34,36, 38		H	H	L	L	H	H	H	H	H	H	
E13		4,7,12,16,32,33	2,3,6,8,9,11,14, 15,19,20,22,23,25, 26,28,30,34,36,37, 38,39		H	H	H	H	L	L	H	H	H	H	
E14		4,7,8,9,16,26,28	2,3,6,11,12,14,15, 19,20,22,23,25,30, 32,33,34,36,37,38, 39		H	H	H	H	H	H	L	L	H	H	
E15		4,6,7,9,11,14,15, 16,19,22,25,30,33, 37,38	2,3,8,12,20,23,26, 28,32,34,36,39		H	H	H	H	H	H	H	H	L	L	
E16		Open	2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,25,26,28,30, 32,33,34,36,37,38, 39		H	H	H	H	H	H	H	H	H	H	
E17		26	2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,25,28,30,32, 33,34,36,37,38,39		L	L	L	L	L	L	L	L	L	L	
E18		2,8,12,20,22,28, 32,34,36,39	3,4,6,7,9,11,14, 15,16,19,23,25,26, 30,33,37,38		H	H	H	H	H	H	H	H	H	H	

See footnotes at end of table.

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TABLE VII. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states 1/										Test figure
		A to pin	B to pin	C to pin	1F0	1F1	1F2	1F3	2F0	2F1	2F2	2F3	3F0	3F1	
E19	Functional	2,8,12,20,22,26, 28,32,34,36,39	3,4,6,7,9,11,14, 15,16,19,23,25,30, 33,37,38	2/, 3/	L	L	L	L	L	L	L	L	L	L	9
E20		3,6,9,11,14,15,19, 23,33,38	2,4,7,8,12,16,20, 22,25,26,28,30,32, 34,36,37,39		H	H	H	H	H	H	H	H	H	H	
E21		3,6,9,11,14,15,19, 23,26,33,38	2,4,7,8,12,16,20, 22,25,28,30,32,34, 36,37,39		L	L	L	L	L	L	L	L	L	L	
E22		2,3,6,8,9,11,12, 14,15,19,20,22,23, 28,32,33,34,36,38, 39	4,7,16,25,26,30,37		H	H	H	H	H	H	H	H	H	H	
E23		2,3,6,8,9,11,12, 14,15,19,20,22,23, 26,28,32,33,34,36, 38,39	4,7,16,25,30,37		L	L	L	L	L	L	L	L	L	L	
E24		2,3,6,8,9,11,12, 14,15,19,20,22,23, 25,28,30,32,33,34, 36,37,38,39	4,7,16,26		H	H	H	H	H	H	H	H	H	H	
E25		2,3,6,8,9,11,12, 14,15,19,20,22,23, 25,26,28,30,32,33, 34,36,37,38,39	4,7,16		H	H	H	H	H	H	H	H	H	H	
E26		3,4,6,7,9,11,14, 15,16,19,23,33,38	2,8,12,20,22,25, 26,28,30,32,34,36, 37,39		H	H	H	H	H	H	H	H	H	H	
E27		3,4,6,7,9,11,14, 15,16,19,23,26,33, 38	2,8,12,20,22,25, 28,30,32,34,36,37, 39		H	H	H	H	H	H	H	H	H	H	

See footnotes at end of table.

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TABLE VII. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states 1/										Test figure
		A to pin	B to pin	C to pin	1F0	1F1	1F2	1F3	2F0	2F1	2F2	2F3	3F0	3F1	
E28	Functional	4,7,16,25,26,30,37	2,3,6,8,9,11,12, 14,15,19,20,22,23, 28,32,33,34,36,38, 39	2/, 3/	L	L	L	L	L	L	L	L	L	L	9
E29		2,4,7,8,16,20,22, 25,26,28,30,32,34, 36,37,39	3,6,9,11,12,14,15, 19,23,33,38		L	L	L	L	L	L	L	L	L	L	
E30		3,4,6,7,9,11,12, 14,15,16,19,23,25, 26,30,33,37,38	2,8,20,22,28,32, 34,36,39		H	H	H	H	H	H	H	H	H	H	
E31		2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,25,28,30,32, 33,34,36,37,38,39	26		H	H	H	H	H	H	H	H	H	H	
E32		2,3,4,6,7,8,9,11, 12,14,15,16,19,20, 22,23,25,26,28,30, 32,33,34,36,37,38, 39	Open		H	H	H	H	H	H	H	H	H	H	

1/ Output logic states:
L = Low-level voltage = 0.0 to 0.3 V.
H = High-level voltage = 2.4 to 4.5 V.

2/ For test conditions E1-E32, verification of the output logic states in footnote 1 is obtained by connecting voltmeter M02, point C, to each output pin in succession after input conditions have been accomplished.

3/ 1F0, 1F1, 1F2, 1F3, 2F0, 2F1, 2F2, 2F3, 3F0, 3F1 output logic states are measured at output pins 18, 40, 35, 17, 29, 13, 5, 27, 21, and 24, respectively.

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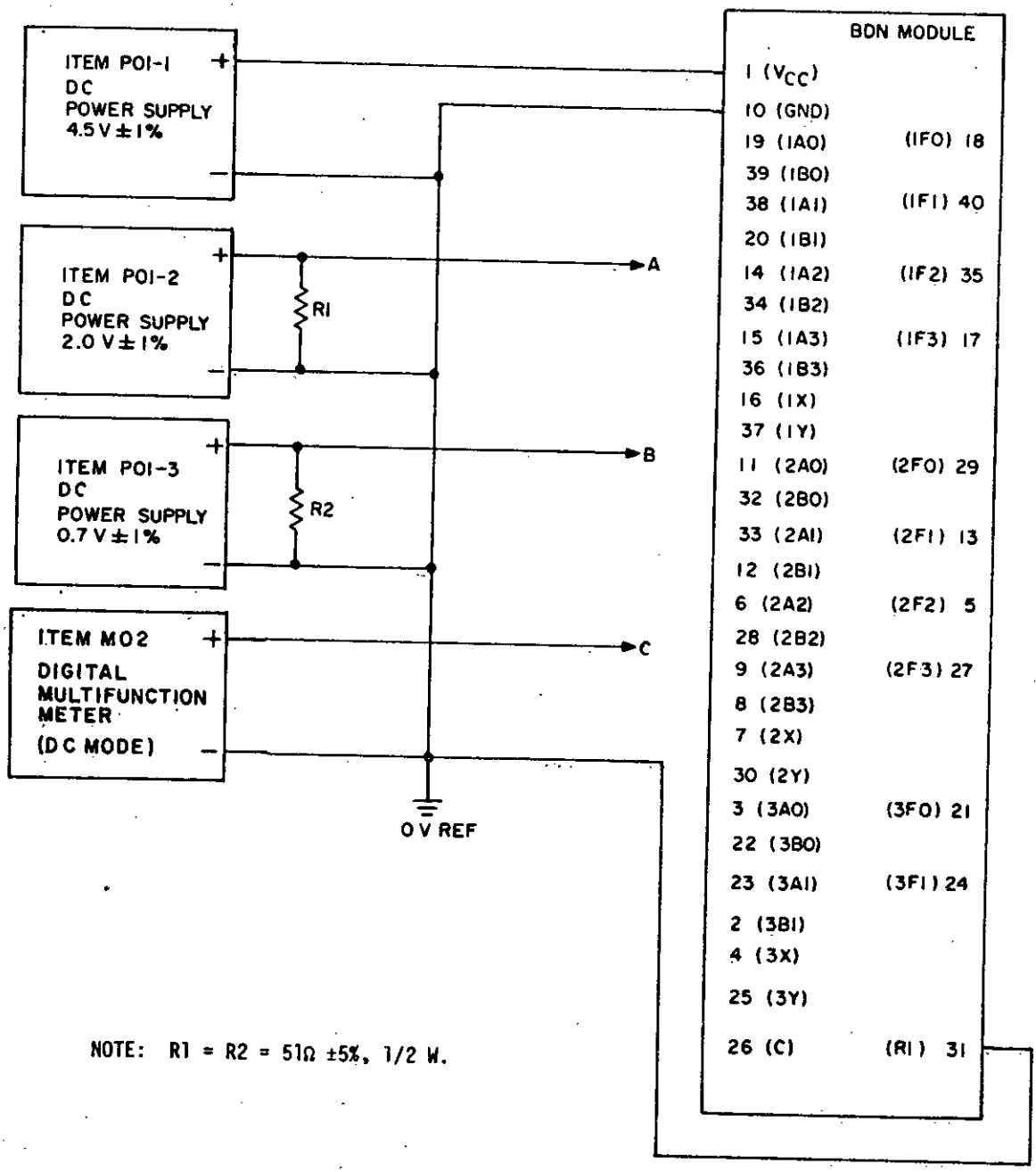


FIGURE 9. Functional test circuit for table VII.

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TABLE VIII. Static output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1F0 Static output voltage, low level	26	Open	18	0.0	0.3	V	10
F2	1F1			40				
F3	1F2			35				
F4	1F3			17				
F5	2F0			29				
F6	2F1			13				
F7	2F2			5				
F8	2F3			27				
F9	3F0			21				
F10	3F1			24				

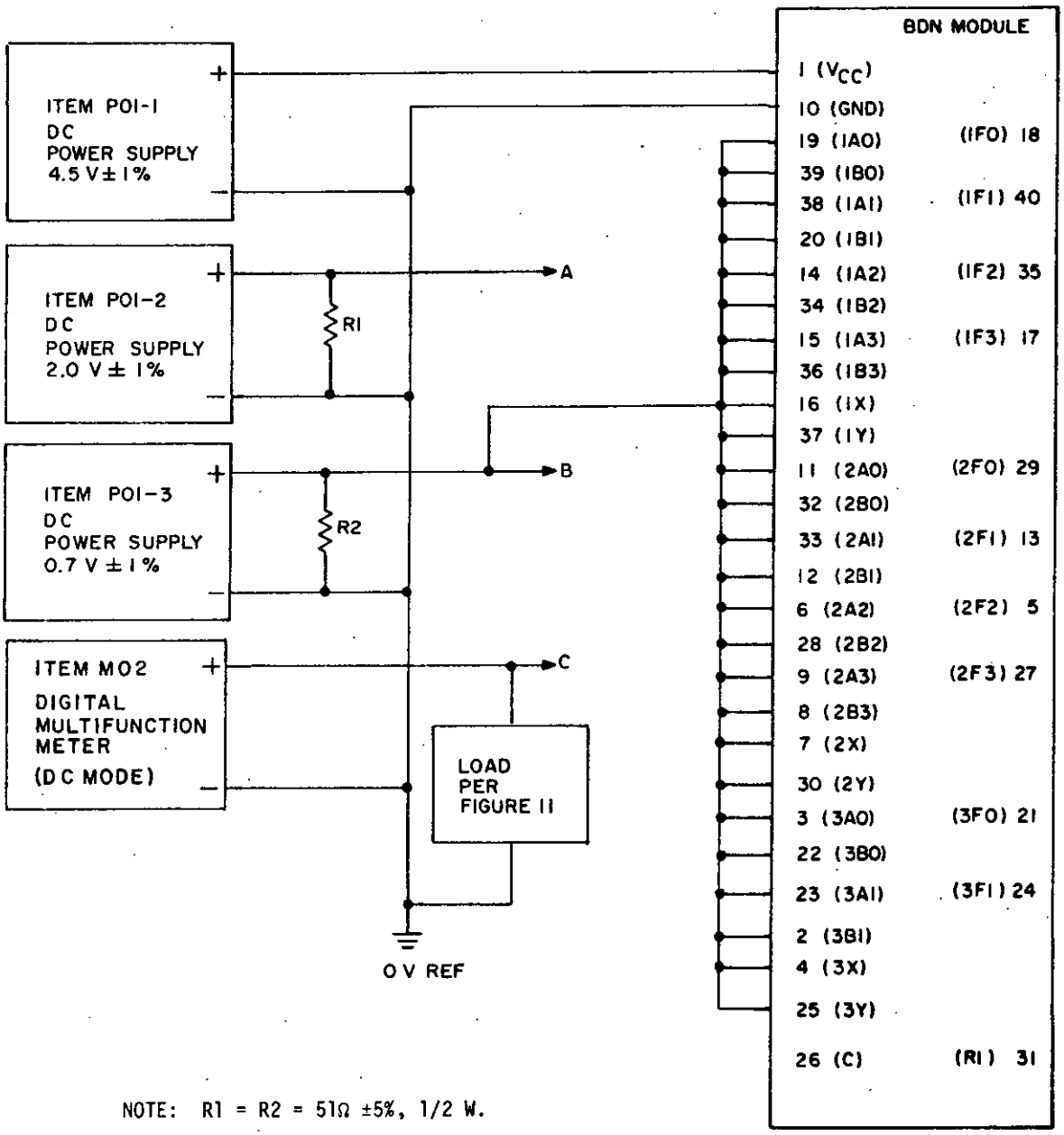
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TABLE IX. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1F0 Static output voltage, high level	Open	26	18	2.4	4.5	V	10
G2	1F1			40				
G3	1F2			35				
G4	1F3			17				
G5	2F0			29				
G6	2F1			13				
G7	2F2			5				
G8	2F3			27				
G9	3F0			21				
G10	3F1			24				

MIL-M-38510/(obsolete)

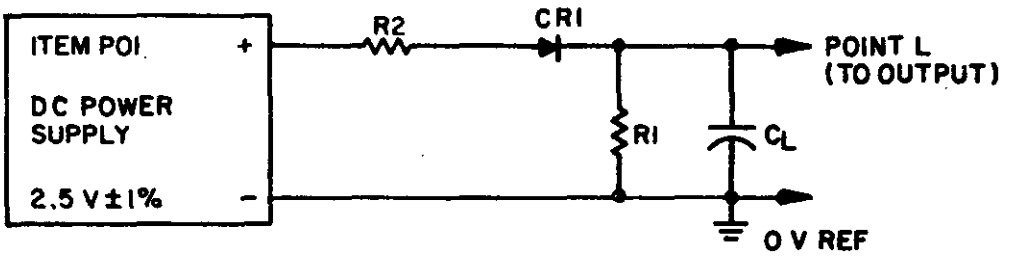
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 10. Static output voltage (low-level and high-level) test circuit for tables VIII and IX.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. R1 = 6.04 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 4.0 mA ±1% out of point L with point L held at 0.3 V.
4. For test conditions L1-L10, M1-M10, N1-N80, Q1-Q80, T1-T10 and U1-U10, C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions L11-L20, M11-M20, P1-P80 and R1-R80, C_L = 150 ±15 pF (including probe and parasitic capacitance).
6. Capacitor C_L is not required for test conditions F1-F10 and G1-G10.

FIGURE 11. BDN logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	1A0 Input current, high level	19	2,3,4,6,7,8,9,11,12,14,15,16, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39	0.0	20.0	µA	12
H2	1A1	38	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,33, 34,36,37,39				
H3	1A2	14	2,3,4,6,7,8,9,11,12,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H4	1A3	15	2,3,4,6,7,8,9,11,12,14,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H5	2A0	11	2,3,4,6,7,8,9,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H6	2A1	33	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,34, 36,37,38,39				
H7	2A2	6	2,3,4,7,8,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H8	2A3	9	2,3,4,6,7,8,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H9	3A0	3	2,4,6,7,8,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				

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TABLE X. Input current (high-level) test - Continued.

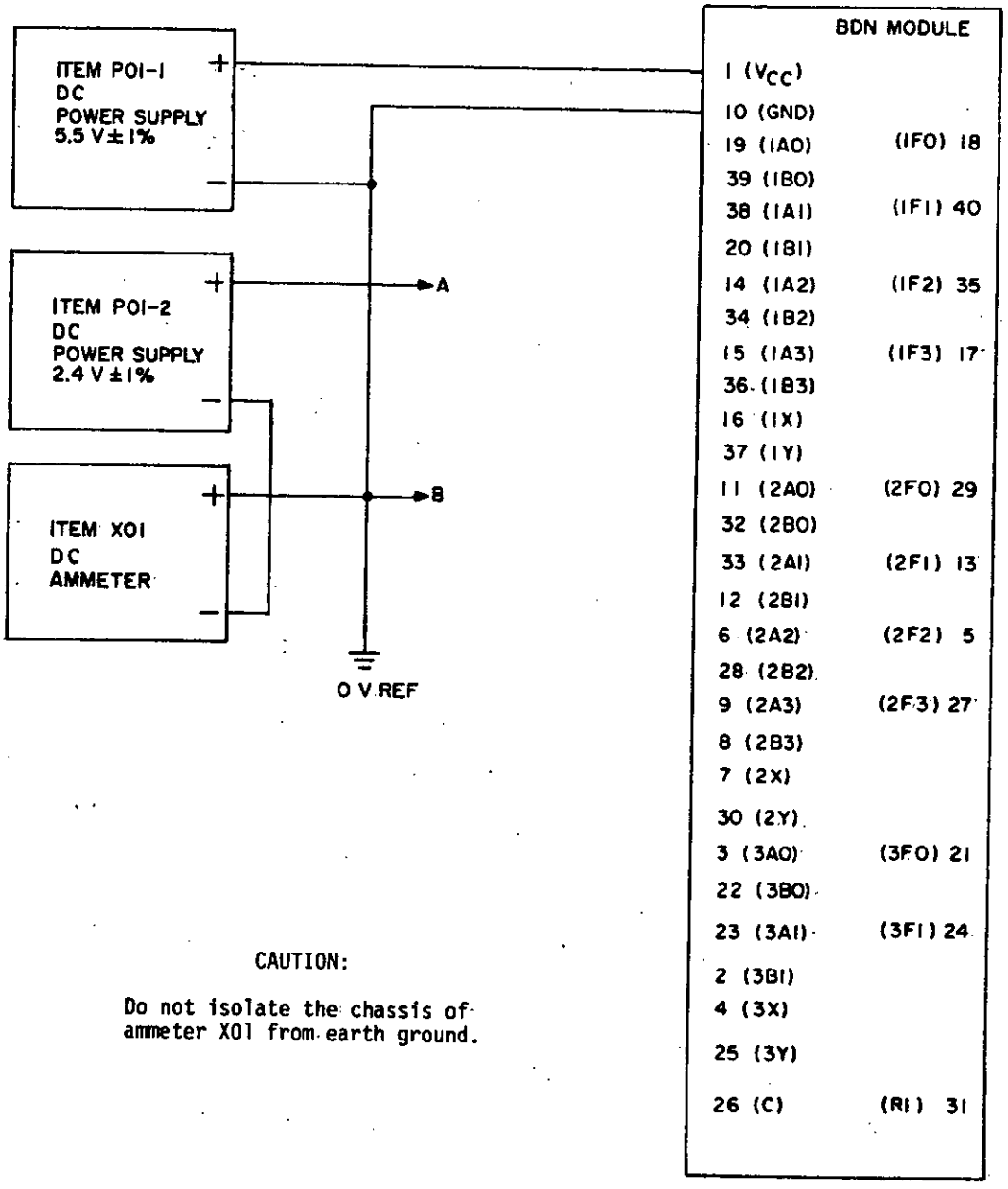
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H10	3A1 Input current, high level	23	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,25,26,28,30,32,33,34, 36,37,38,39	0.0	20.0	μA	12
H11	3X	4	2,3,6,7,8,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H12	3Y	25	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,26,28,30,32,33,34, 36,37,38,39				
H13	C	26	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,28,30,32,33,34, 36,37,38,39				
H14	1B0	39	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,33, 34,36,37,38		30.0		
H15	1B1	20	2,3,4,6,7,8,9,11,12,14,15,16, 19,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H16	1B2	34	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,33, 36,37,38,39				
H17	1B3	36	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,33, 34,37,38,39				
H18	2B0	32	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,33,34, 36,37,38,39				

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H19	2B1 Input current, high level	12	2,3,4,6,7,8,9,11,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39	0.0	30.0	µA	12
H20	2B2	28	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,30,32,33,34, 36,37,38,39				
H21	2B3	8	2,3,4,6,7,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H22	3B0	22	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,23,25,26,28,30,32,33,34, 36,37,38,39				
H23	3B1	2	3,4,6,7,8,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H24	1X	16	2,3,4,6,7,8,9,11,12,14,15,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39		40.0		
H25	1Y	37	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,30,32,33, 34,36,38,39				
H26	2X	7	2,3,4,6,8,9,11,12,14,15,16,19, 20,22,23,25,26,28,30,32,33,34, 36,37,38,39				
H27	2Y	30	2,3,4,6,7,8,9,11,12,14,15,16, 19,20,22,23,25,26,28,32,33,34, 36,37,38,39				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 12. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

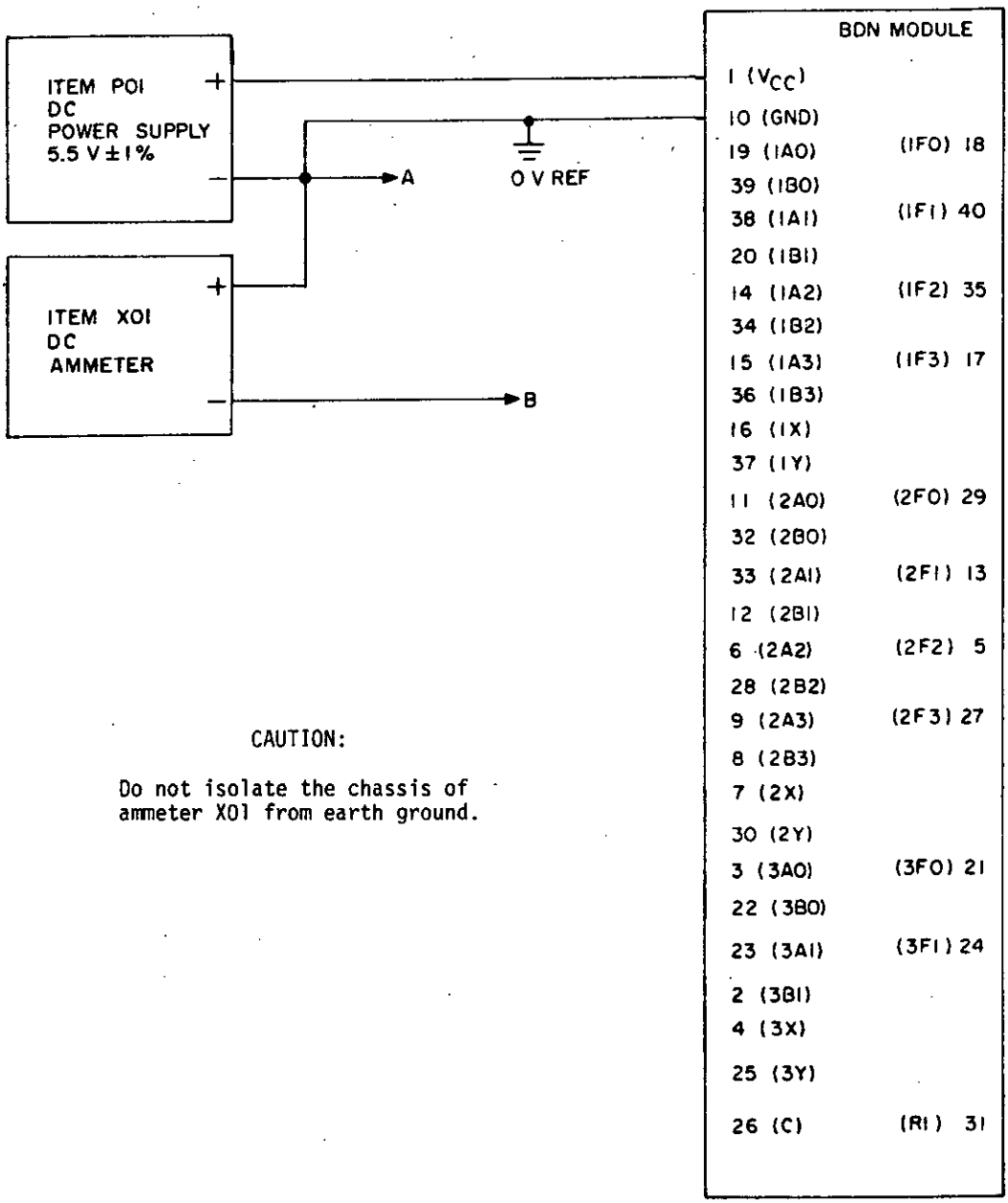
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A0 Input current, low level	Open	19	-0.10	-0.43	mA	13
J2	1A1		38				
J3	1A2		14				
J4	1A3		15				
J5	2A0		11				
J6	2A1		33				
J7	2A2		6				
J8	2A3		9				
J9	3A0		3				
J10	3A1		23				
J11	3X		4				
J12	3Y		25				
J13	C		26		-0.40		
J14	1B0	16	39		-0.63		
J15	1B1		20				
J16	1B2		34				
J17	1B3		36				
J18	2B0	7	32				
J19	2B1	7	12				

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J20	2B2 Input current, low level	7	28	-0.10	-0.63	mA	13
J21	2B3	7	8				
J22	3B0	4	22				
J23	3B1	4	2				
J24	1X	Open	16	-0.20	-0.86		
J25	1Y		37				
J26	2X		7				
J27	2Y		30				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Input current (low-level) test circuit for table XI.

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TABLE XII. Power supply current (inputs at low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply current (inputs at high level)	C	Open	20	64	mA	14
K2	Power supply current (inputs at low level)	Open	C	18	60	mA	14

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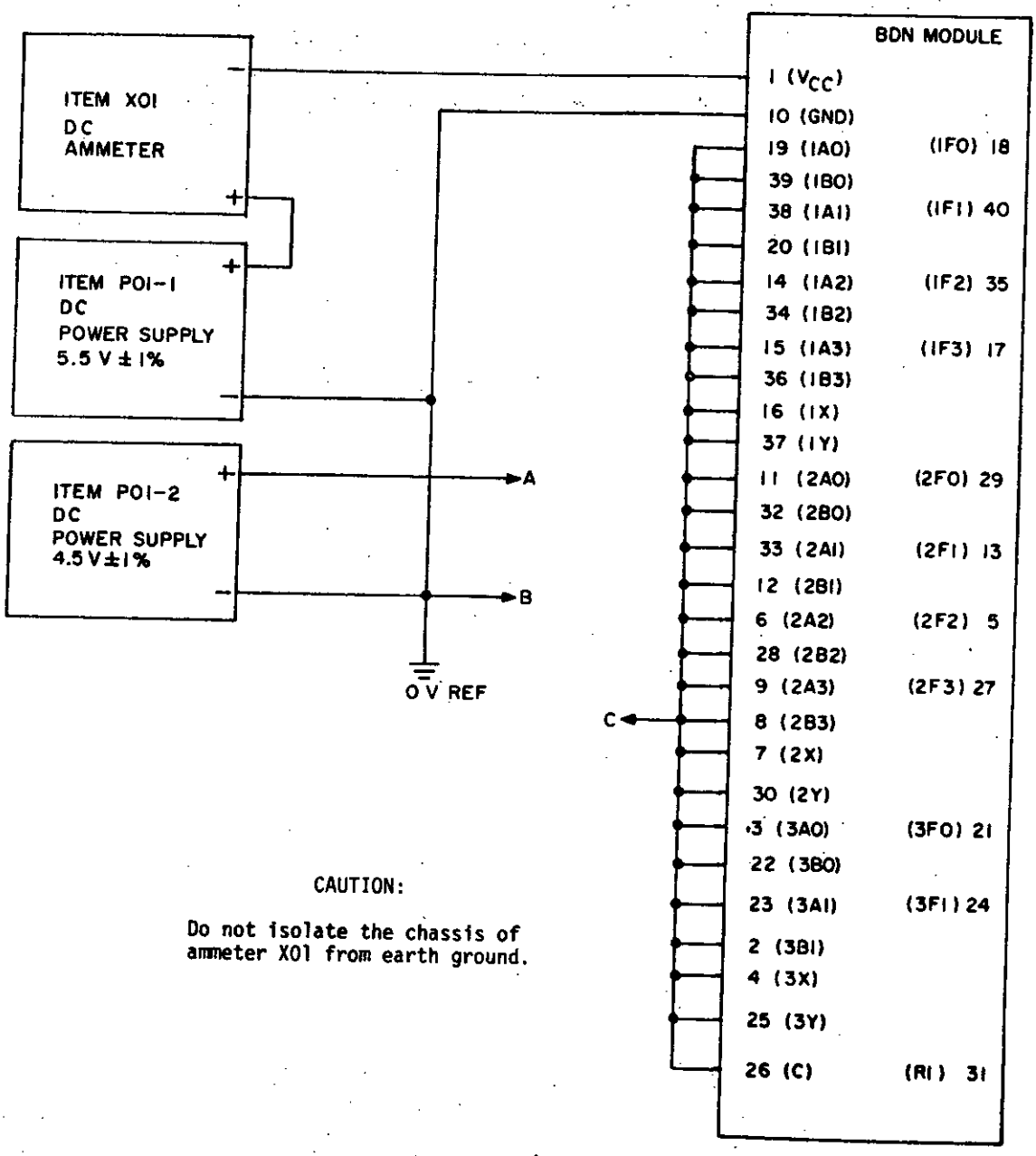


FIGURE 14. Power supply current (inputs at low-level and high-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1F0 Transition time (t_{TLH}), C_L = 50 pF	16,37	19	Open	18	2	20	ns	15,16
L2	1F1		38		40				
L3	1F2		14		35				
L4	1F3		15		17				
L5	2F0	7,30	11		29				
L6	2F1		33		13				
L7	2F2		6		5				
L8	2F3		9		27				
L9	3F0	4,25	3		21				
L10	3F1	4,25	23		24				
L11	1F0 Transition time (t_{TLH}), C_L = 150 pF	16,37	19		18		40		
L12	1F1		38		40				
L13	1F2		14		35				
L14	1F3		15		17				
L15	2F0	7,30	11		29				
L16	2F1	7,30	33		13				

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TABLE XIII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L17	2F2 Transition time (t_{TLH}), C_L = 150 pF	7,30	6	Open	5	2	40	ns	15,16
L18	2F3	7,30	9		27				
L19	3F0	4,25	3		21				
L20	3F1	4,25	23		24				

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TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
M1	1F0 Transition time (t_{THL}), C_L = 50 pF	16,37	19	Open	18	2	15	ns	15,16
M2	1F1		38		40				
M3	1F2		14		35				
M4	1F3		15		17				
M5	2F0	7,30	11		29				
M6	2F1		33		13				
M7	2F2		6		5				
M8	2F3		9		27				
M9	3F0	4,25	3		21				
M10	3F1	4,25	23		24				
M11	1F0 Transition time (t_{THL}), C_L = 150 pF	16,37	19		18		30		
M12	1F1		38		40				
M13	1F2		14		35				
M14	1F3		15		17				
M15	2F0	7,30	11		29				
M16	2F1	7,30	33		13				

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TABLE XIV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M17	2F2 Transition time (t_{THL}), C_L = 150 pF	7,30	6	Open	5	2	30	ns	15,16
M18	2F3	7,30	9		27				
M19	3F0	4,25	3		21				
M20	3F1	4,25	23		24				

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TABLE XV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1A0 to 1F0 Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	16,37	19	Open	18	10	70	ns	15,16
N2	1A1 to 1F1		38		40				
N3	1A2 to 1F2		14		35				
N4	1A3 to 1F3		15		17				
N5	2A0 to 2F0	7,30	11		29				
N6	2A1 to 2F1		33		13				
N7	2A2 to 2F2		6		5				
N8	2A3 to 2F3		9		27				
N9	3A0 to 3F0	4,25	3		21				
N10	3A1 to 3F1	4,25	23		24				
N11	1B0 to 1F0	37	39	16	18				
N12	1B1 to 1F1		20		40				
N13	1B2 to 1F2		34		35				
N14	1B3 to 1F3		36		17				
N15	2B0 to 2F0	30	32	7	29				
N16	2B1 to 2F1	30	12	7	13				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N17	2B2 to 2F2 Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	30	28	7	5	10	70	ns	15,16
N18	2B3 to 2F3	30	8	7	27				
N19	3B0 to 3F0	25	22	4	21				
N20	3B1 to 3F1	25	2	4	24				
N21	1B0 to 1F0 Propagation delay time (t_{PHL}), inverting, C_L = 50 pF	16	39	37	18	20	110		15,17
N22	1B1 to 1F1		20		40				
N23	1B2 to 1F2		34		35				
N24	1B3 to 1F3		36		17				
N25	2B0 to 2F0	7	32	30	29				
N26	2B1 to 2F1		12		13				
N27	2B2 to 2F2		28		5				
N28	2B3 to 2F3		8		27				
N29	3B0 to 3F0	4	22	25	21				
N30	3B1 to 3F1	4	2	25	24				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N31	1X to 1F0	Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	37,19	16	39	18	20	70	ns	15,16
N32	1X to 1F1		37,38		20	40				
N33	1X to 1F2		37,14		34	35				
N34	1X to 1F3		37,15		36	17				
N35	2X to 2F0		30,11	7	32	29				
N36	2X to 2F1		30,33		12	13				
N37	2X to 2F2		30,6		28	5				
N38	2X to 2F3		30,9		8	27				
N39	3X to 3F0		25,3	4	22	21				
N40	3X to 3F1		25,23	4	2	24				
N41	1X to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 50 pF	39	16	26,37	18	15			15,17
N42	1X to 1F1		20			40				
N43	1X to 1F2		34			35				
N44	1X to 1F3		36			17				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N45	2X to 2F0	Propagation delay time (t_{PHL}), inverting, C_L = 50 pF	32	7	26,30	29	15	70	ns	15,17
N46	2X to 2F1		12			13				
N47	2X to 2F2		28			5				
N48	2X to 2F3		8			27				
N49	3X to 3F0		22	4	26,25	21				
N50	3X to 3F1		2	4	26,25	24				
N51	1Y to 1F0	Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	26,39	37	16	18	20			15,16
N52	1Y to 1F1		26,20			40				
N53	1Y to 1F2		26,34			35				
N54	1Y to 1F3		26,36			17				
N55	2Y to 2F0		26,32	30	7	29				
N56	2Y to 2F1		26,12			13				
N57	2Y to 2F2		26,28			5				
N58	2Y to 2F3		26,8			27				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N59	3Y to 3F0	Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	26,22	25	4	21	20	70	ns	15,16
N60	3Y to 3F1	Propagation delay time (t_{PHL}), noninverting, C_L = 50 pF	26,2	25	4	24	20			15,16
N61	1Y to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 50 pF	16	37	19,39	18	15			15,17
N62	1Y to 1F1				38,20	40				
N63	1Y to 1F2				14,34	35				
N64	1Y to 1F3				15,36	17				
N65	2Y to 2F0		7	30	11,32	29				
N66	2Y to 2F1				33,12	13				
N67	2Y to 2F2				6,28	5				
N68	2Y to 2F3				9,8	27				
N69	3Y to 3F0		4	25	3,22	21				
N70	3Y to 3F1		4	25	23,2	24				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N71	C to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 50 pF	Open	26	16,37	18	20	115	ns	15,17
N72	C to 1F1					40				
N73	C to 1F2					35				
N74	C to 1F3					17				
N75	C to 2F0				7,30	29				
N76	C to 2F1					13				
N77	C to 2F2					5				
N78	C to 2F3					27				
N79	C to 3F0				4,25	21				
N80	C to 3F1				4,25	24				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1A0 to 1F0 Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	16,37	19	Open	18	10	85	ns	15,16
P2	1A1 to 1F1		38		40				
P3	1A2 to 1F2		14		35				
P4	1A3 to 1F3		15		17				
P5	2A0 to 2F0	7,30	11		29				
P6	2A1 to 2F1		33		13				
P7	2A2 to 2F2		6		5				
P8	2A3 to 2F3		9		27				
P9	3A0 to 3F0	4,25	3		21				
P10	3A1 to 3F1	4,25	23		24				
P11	1B0 to 1F0	37	39	16	18				
P12	1B1 to 1F1		20		40				
P13	1B2 to 1F2		34		35				
P14	1B3 to 1F3		36		17				
P15	2B0 to 2F0	30	32	7	29				
P16	2B1 to 2F1	30	12	7	13				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P17	2B2 to 2F2 Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	30	28	7	5	10	85	ns	15,16
P18	2B3 to 2F3	30	8	7	27				
P19	3B0 to 3F0	25	22	4	21				
P20	3B1 to 3F1	25	2	4	24				
P21	1B0 to 1F0 Propagation delay time (t_{PHL}), inverting, C_L = 150 pF	16	39	37	18	20	125		15,17
P22	1B1 to 1F1		20		40				
P23	1B2 to 1F2		34		35				
P24	1B3 to 1F3		36		17				
P25	2B0 to 2F0	7	32	30	29				
P26	2B1 to 2F1		12		13				
P27	2B2 to 2F2		28		5				
P28	2B3 to 2F3		8		27				
P29	3B0 to 3F0	4	22	25	21				
P30	3B1 to 3F1	4	2	25	24				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P31	1X to 1F0	Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	37,19	16	39	18	20	85	ns	15,16
P32	1X to 1F1		37,38		20	40				
P33	1X to 1F2		37,14		34	35				
P34	1X to 1F3		37,15		36	17				
P35	2X to 2F0		30,11	7	32	29				
P36	2X to 2F1		30,33		12	13				
P37	2X to 2F2		30,6		28	5				
P38	2X to 2F3		30,9		8	27				
P39	3X to 3F0		25,3	4	22	21				
P40	3X to 3F1		25,23	4	2	24				
P41	1X to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 150 pF	39	16	26,37	18	15			15,17
P42	1X to 1F1		20			40				
P43	1X to 1F2		34			35				
P44	1X to 1F3		36			17				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
P45	2X to 2F0	Propagation delay time (t_{PHL}), inverting, C_L = 150 pF	32	7	26,30	29	15	85	ns	15,17
P46	2X to 2F1		12			13				
P47	2X to 2F2		28			5				
P48	2X to 2F3		8			27				
P49	3X to 3F0		22	4	26,25	21				
P50	3X to 3F1		2	4	26,25	24				
P51	1Y to 1F0	Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	26,39	37	16	18	20			15,16
P52	1Y to 1F1		26,20			40				
P53	1Y to 1F2		26,34			35				
P54	1Y to 1F3		26,36			17				
P55	2Y to 2F0		26,32	30	7	29				
P56	2Y to 2F1		26,12			13				
P57	2Y to 2F2		26,28			5				
P58	2Y to 2F3		26,8			27				

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P59	3Y to 3F0	Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	26,22	25	4	21	20	85	ns	15,16
P60	3Y to 3F1	Propagation delay time (t_{PHL}), noninverting, C_L = 150 pF	26,2	25	4	24	20			15,16
P61	1Y to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 150 pF	16	37	19,39	18	15			15,17
P62	1Y to 1F1				38,20	40				
P63	1Y to 1F2				14,34	35				
P64	1Y to 1F3				15,36	17				
P65	2Y to 2F0		7	30	11,32	29				
P66	2Y to 2F1				33,12	13				
P67	2Y to 2F2				6,28	5				
P68	2Y to 2F3				9,8	27				
P69	3Y to 3F0		4	25	3,22	21				
P70	3Y to 3F1		4	25	23,2	24				

MIL-M-38510/5B(USAF)

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TABLE XV. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P71	C to 1F0	Propagation delay time (t_{PHL}), inverting, C_L = 150 pF	Open	26	16,37	18	20	125	ns	15,17
P72	C to 1F1					40				
P73	C to 1F2					35				
P74	C to 1F3					17				
P75	C to 2F0				7,30	29				
P76	C to 2F1					13				
P77	C to 2F2					5				
P78	C to 2F3					27				
P79	C to 3F0				4,25	21				
P80	C to 3F1				4,25	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1A0 to 1F0 Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	16,37	19	Open	18	10	75	ns	15,16
Q2	1A1 to 1F1		38		40				
Q3	1A2 to 1F2		14		35				
Q4	1A3 to 1F3		15		17				
Q5	2A0 to 2F0	7,30	11		29				
Q6	2A1 to 2F1		33		13				
Q7	2A2 to 2F2		6		5				
Q8	2A3 to 2F3		9		27				
Q9	3A0 to 3F0	4,25	3		21				
Q10	3A1 to 3F1	4,25	23		24				
Q11	1B0 to 1F0	37	39	16	18				
Q12	1B1 to 1F1		20		40				
Q13	1B2 to 1F2		34		35				
Q14	1B3 to 1F3		36		17				
Q15	2B0 to 2F0	30	32	7	29				
Q16	2B1 to 2F1	30	12	7	13				

MIL-M-38510/13A(EC)

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
Q17	2B2 to 2F2 Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	30	28	7	5	10	75	ns	15,16
Q18	2B3 to 2F3	30	8	7	27				
Q19	3B0 to 3F0	25	22	4	21				
Q20	3B1 to 3F1	25	2	4	24				
Q21	1B0 to 1F0 Propagation delay time (t_{PLH}), inverting, C_L = 50 pF	16	39	37	18	20	115		15,17
Q22	1B1 to 1F1		20		40				
Q23	1B2 to 1F2		34		35				
Q24	1B3 to 1F3		36		17				
Q25	2B0 to 2F0	7	32	30	29				
Q26	2B1 to 2F1		12		13				
Q27	2B2 to 2F2		28		5				
Q28	2B3 to 2F3		8		27				
Q29	3B0 to 3F0	4	22	25	21				
Q30	3B1 to 3F1	4	2	25	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q31	1X to 1F0	Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	37,19	16	39	18	20	75	ns	15,16
Q32	1X to 1F1		37,38		20	40				
Q33	1X to 1F2		37,14		34	35				
Q34	1X to 1F3		37,15		36	17				
Q35	2X to 2F0		30,11	7	32	29				
Q36	2X to 2F1		30,33		12	13				
Q37	2X to 2F2		30,6		28	5				
Q38	2X to 2F3		30,9		8	27				
Q39	3X to 3F0		25,3	4	22	21				
Q40	3X to 3F1		25,23	4	2	24				
Q41	1X to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 50 pF	39	16	26,37	18	15	70		15,17
Q42	1X to 1F1		20			40				
Q43	1X to 1F2		34			35				
Q44	1X to 1F3		36			17				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q45	2X to 2F0	Propagation delay time (t_{PLH}), inverting, C_L = 50 pF	32	7	26,30	29	15	70	ns	15,17
Q46	2X to 2F1		12			13				
Q47	2X to 2F2		28			5				
Q48	2X to 2F3		8			27				
Q49	3X to 3F0		22	4	26,25	21				
Q50	3X to 3F1		2	4	26,25	24				
Q51	1Y to 1F0	Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	26,39	37	16	18	20	75		15,16
Q52	1Y to 1F1		26,20			40				
Q53	1Y to 1F2		26,34			35				
Q54	1Y to 1F3		26,36			17				
Q55	2Y to 2F0		26,32	30	7	29				
Q56	2Y to 2F1		26,12			13				
Q57	2Y to 2F2		26,28			5				
Q58	2Y to 2F3		26,8			27				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q59	3Y to 3F0	Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	26,22	25	4	21	20	75	ns	15,16
Q60	3Y to 3F1	Propagation delay time (t_{PLH}), noninverting, C_L = 50 pF	26,2	25	4	24	20	75		15,16
Q61	1Y to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 50 pF	16	37	19,39	18	15	70		15,17
Q62	1Y to 1F1				38,20	40				
Q63	1Y to 1F2				14,34	35				
Q64	1Y to 1F3				15,36	17				
Q65	2Y to 2F0		7	30	11,32	29				
Q66	2Y to 2F1				33,12	13				
Q67	2Y to 2F2				6,28	5				
Q68	2Y to 2F3				9,8	27				
Q69	3Y to 3F0		4	25	3,22	21				
Q70	3Y to 3F1		4	25	23,2	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q71	C to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 50 pF	Open	26	16,37	18	20	115	ns	15,17
Q72	C to 1F1					40				
Q73	C to 1F2					35				
Q74	C to 1F3					17				
Q75	C to 2F0				7,30	29				
Q76	C to 2F1					13				
Q77	C to 2F2					5				
Q78	C to 2F3					27				
Q79	C to 3F0				4,25	21				
Q80	C to 3F1				4,25	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1A0 to 1F0 Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	16,37	19	Open	18	10	90	ns	15,16
R2	1A1 to 1F1		38		40				
R3	1A2 to 1F2		14		35				
R4	1A3 to 1F3		15		17				
R5	2A0 to 2F0	7,30	11		29				
R6	2A1 to 2F1		33		13				
R7	2A2 to 2F2		6		5				
R8	2A3 to 2F3		9		27				
R9	3A0 to 3F0	4,25	3		21				
R10	3A1 to 3F1	4,25	23		24				
R11	1B0 to 1F0	37	39	16	18				
R12	1B1 to 1F1		20		40				
R13	1B2 to 1F2		34		35				
R14	1B3 to 1F3		36		17				
R15	2B0 to 2F0	30	32	7	29				
R16	2B1 to 2F1	30	12	7	13				

MIL-M-28787/23(EC)

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R17	2B2 to 2F2 Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	30	28	7	5	10	90	ns	15,16
R18	2B3 to 2F3	30	8	7	27				
R19	3B0 to 3F0	25	22	4	21				
R20	3B1 to 3F1	25	2	4	24				
R21	1B0 to 1F0 Propagation delay time (t_{PLH}), inverting, C_L = 150 pF	16	39	37	18	20	130		15,17
R22	1B1 to 1F1		20		40				
R23	1B2 to 1F2		34		35				
R24	1B3 to 1F3		36		17				
R25	2B0 to 2F0	7	32	30	29				
R26	2B1 to 2F1		12		13				
R27	2B2 to 2F2		28		5				
R28	2B3 to 2F3		8		27				
R29	3B0 to 3F0	4	22	25	21				
R30	3B1 to 3F1	4	2	25	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R31	1X to 1F0	Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	37,19	16	39	18	20	90	ns	15,16
R32	1X to 1F1		37,38		20	40				
R33	1X to 1F2		37,14		34	35				
R34	1X to 1F3		37,15		36	17				
R35	2X to 2F0		30,11	7	32	29				
R36	2X to 2F1		30,33		12	13				
R37	2X to 2F2		30,6		28	5				
R38	2X to 2F3		30,9		8	27				
R39	3X to 3F0		25,3	4	22	21				
R40	3X to 3F1		25,23	4	2	24				
R41	1X to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 150 pF	39	16	26,37	18	15	85		15,17
R42	1X to 1F1		20			40				
R43	1X to 1F2		34			35				
R44	1X to 1F3		36			17				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R45	2X to 2F0	Propagation delay time (t_{PLH}), inverting, C_L = 150 pF	32	7	26,30	29	15	85	ns	15,17
R46	2X to 2F1		12			13				
R47	2X to 2F2		28			5				
R48	2X to 2F3		8			27				
R49	3X to 3F0		22	4	26,25	21				
R50	3X to 3F1		2	4	26,25	24				
R51	1Y to 1F0	Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	26,39	37	16	18	20	90		15,16
R52	1Y to 1F1		26,20			40				
R53	1Y to 1F2		26,34			35				
R54	1Y to 1F3		26,36			17				
R55	2Y to 2F0		26,32	30	7	29				
R56	2Y to 2F1		26,12			13				
R57	2Y to 2F2		26,28			5				
R58	2Y to 2F3		26,8			27				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

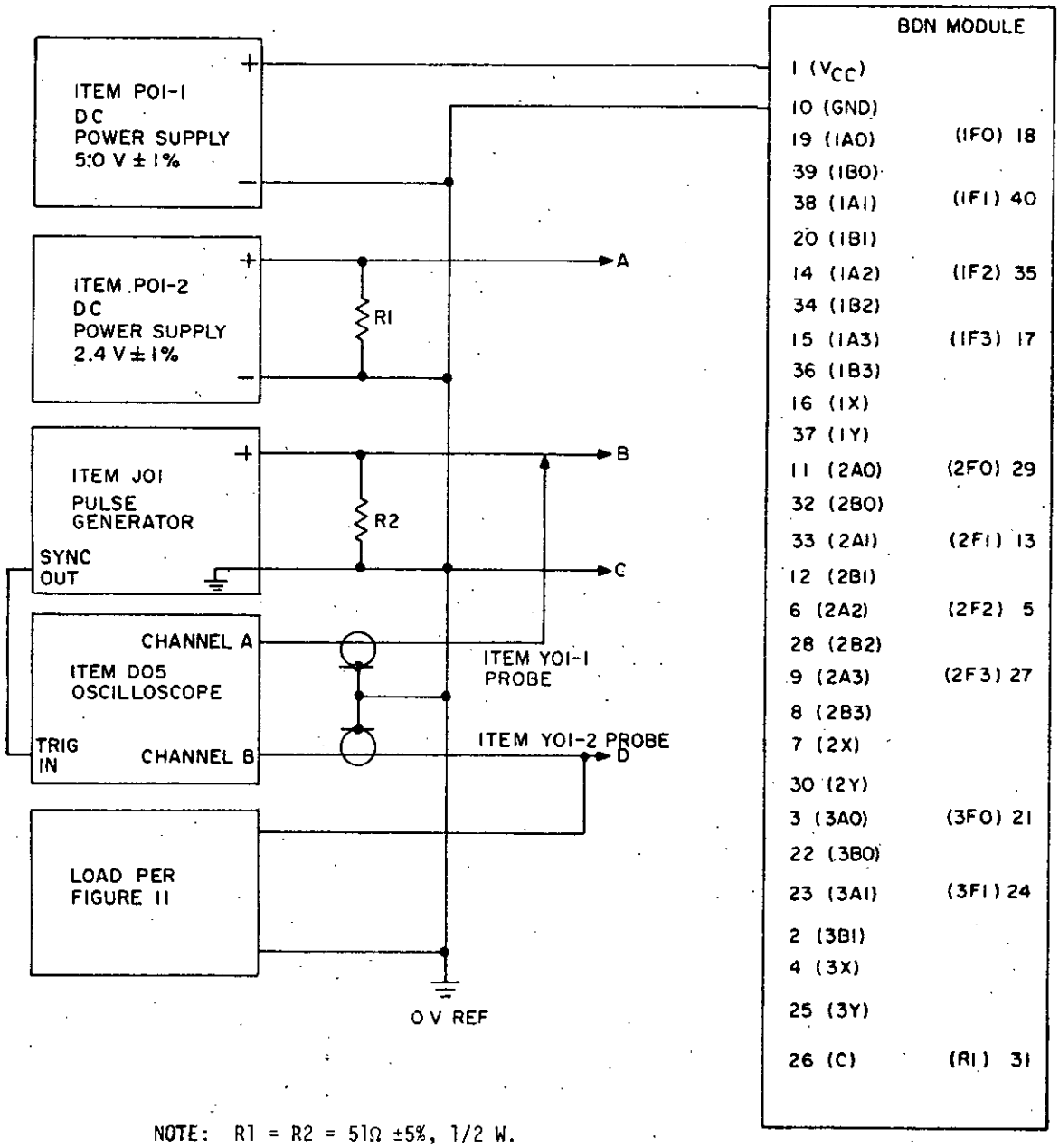
Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R59	3Y to 3F0	Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	26,22	25	4	21	20	90	ns	15,16
R60	3Y to 3F1	Propagation delay time (t_{PLH}), noninverting, C_L = 150 pF	26,2	25	4	24	20	90		15,16
R61	1Y to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 150 pF	16	37	19,39	18	15	85		15,17
R62	1Y to 1F1				38,20	40				
R63	1Y to 1F2				14,34	35				
R64	1Y to 1F3				15,36	17				
R65	2Y to 2F0		7	30	11,32	29				
R66	2Y to 2F1				33,12	13				
R67	2Y to 2F2				6,28	5				
R68	2Y to 2F3				9,8	27				
R69	3Y to 3F0		4	25	3,22	21				
R70	3Y to 3F1		4	25	23,2	24				

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TABLE XVI. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R71	C to 1F0	Propagation delay time (t_{PLH}), inverting, C_L = 150 pF	Open	26	16,37	18	20	125	ns	15,17
R72	C to 1F1					40				
R73	C to 1F2					35				
R74	C to 1F3					17				
R75	C to 2F0				7,30	29				
R76	C to 2F1					13				
R77	C to 2F2					5				
R78	C to 2F3					27				
R79	C to 3F0				4,25	21				
R80	C to 3F1				4,25	24				

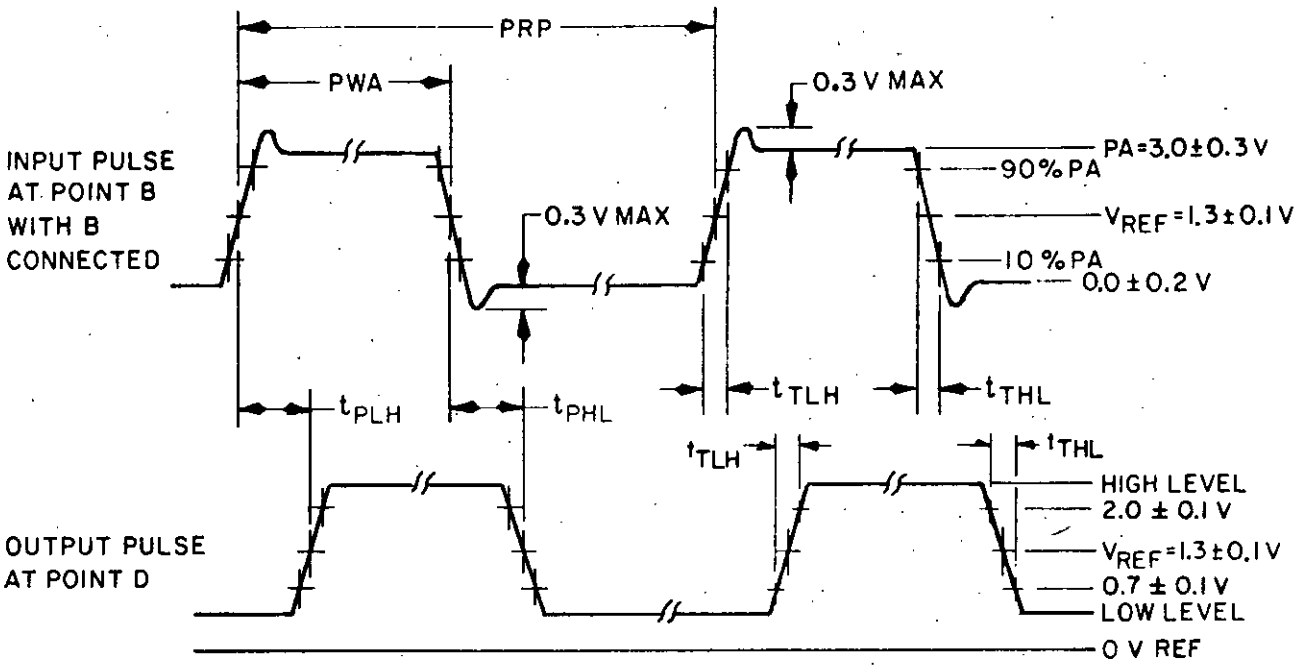
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XIII, XIV, XV, and XVI.

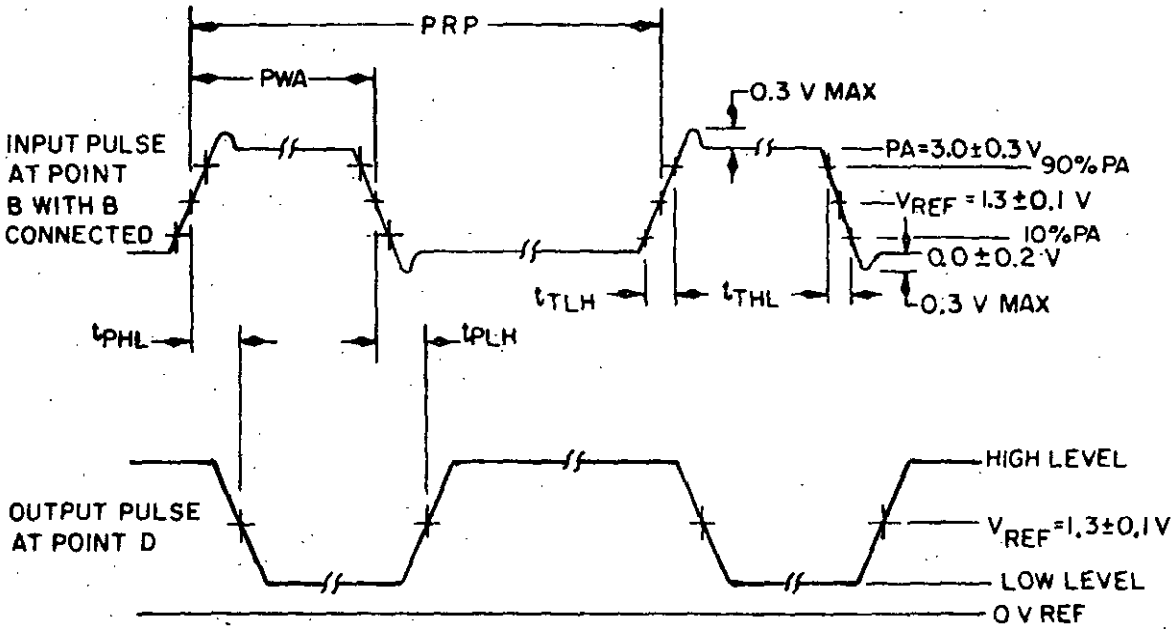
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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   PWA = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
2. Before making the measurements specified in tables XIII, XIV, XV, and XVI, verify that the output pulse low level is $\leq$ 0.3 V and the high level is $\geq$ 2.4 V.
3. All voltages are referenced to ground.

FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XIII, XIV, XV, and XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PWA = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
2. Before making the measurements specified in tables XV and XVI verify that the output pulse low level is $\leq$ 0.3 V and the high level is $\geq$ 2.4 V.
3. All voltages are referenced to ground.

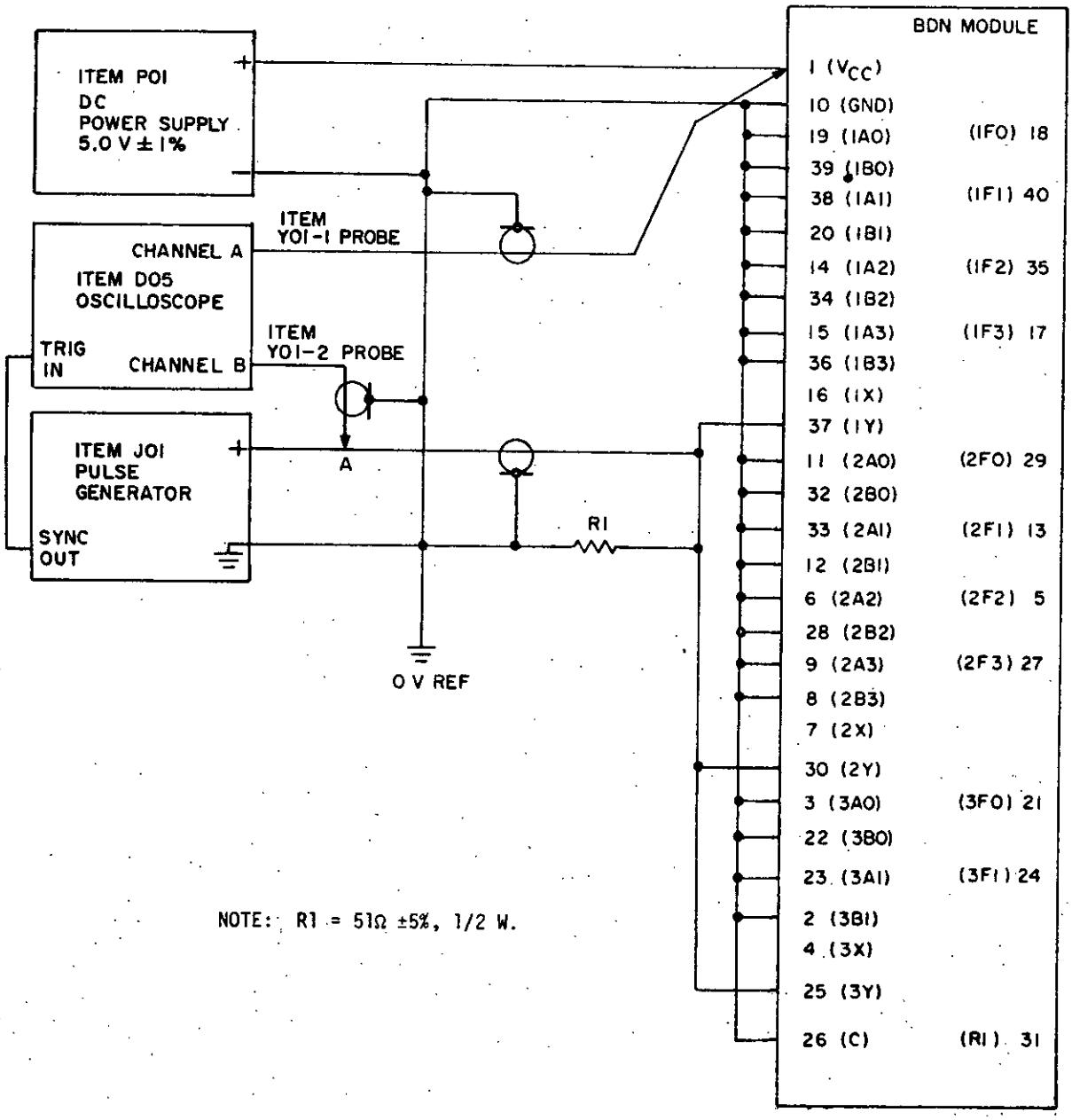
FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XV and XVI.

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TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	18,19

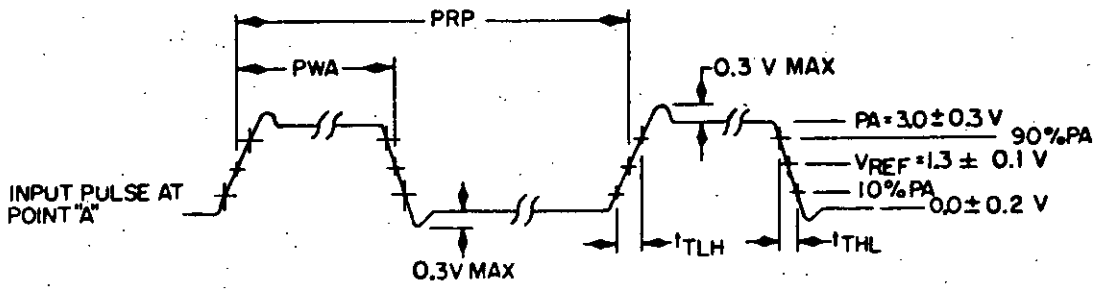
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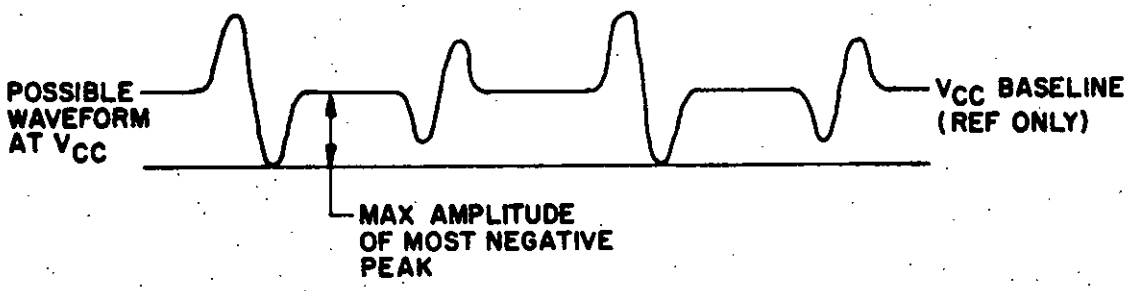


NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 18. Power supply transient amplitude test circuit for table XVII.

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NOTE: Input pulse parameters:
$t_{TLH} = t_{THL} = 10 \pm 2$ ns.
PWA = 500 ±50 ns.
PRP = 1.0 ±0.1 µs.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1F0 Output crosstalk, low level	39	2,3,6,8,9,11,12,14, 15,19,20,22,23,28, 32,33,34,36,38	25,30, 37	18	0	400	mV	20,21
T2	1F1	20	2,3,6,8,9,11,12,14, 15,19,22,23,28,32, 33,34,36,38,39		40				
T3	1F2	34	2,3,6,8,9,11,12,14, 15,19,20,22,23,28, 32,33,36,38,39		35				
T4	1F3	36	2,3,6,8,9,11,12,14, 15,19,20,22,23,28, 32,33,34,38,39		17				
T5	2F0	32	2,3,6,8,9,11,12,14, 15,19,20,22,23,28, 33,34,36,38,39		29				
T6	2F1	12	2,3,6,8,9,11,14,15, 19,20,22,23,28,32, 33,34,36,38,39		13				
T7	2F2	28	2,3,6,8,9,11,12,14, 15,19,20,22,23,32, 33,34,36,38,39		5				
T8	2F3	8	2,3,6,9,11,12,14, 15,19,20,22,23,28, 32,33,34,36,38,39		27				
T9	3F0	22	2,3,6,8,9,11,12,14, 15,19,20,23,28,32, 33,34,36,38,39		21				

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TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T10	3F1 Output crosstalk, low level	2	3,6,8,9,11,12,14, 15,19,20,22,23,28, 32,33,34,36,38,39	25,30, 37	24	0	400	mV	20,21

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TABLE XIX. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1F0 Output crosstalk, high level	14,15, 19,38	2,3,6,8,9,11, 12,22,23,28,32, 33,37,39	20,25,30, 34,36	18	0	400	mV	20,21
U2	1F1		2,3,6,8,9,11, 12,20,22,23,28, 32,33,37	39,25,30, 34,36	40				
U3	1F2		2,3,6,8,9,11, 12,22,23,28,32, 33,34,37	20,39,36, 30,25	35				
U4	1F3		2,3,6,8,9,11, 12,22,23,28,32, 33,36,37	20,39,34, 30,25	17				
U5	2F0	6,9, 11,33	2,3,14,15,19, 20,22,23,30,32, 34,36,38,39	37,12,8, 28,25	29				
U6	2F1		2,3,12,14,15, 19,20,22,23,30, 34,36,38,39	37,32,8, 28,25	13				
U7	2F2		2,3,14,15,19, 20,22,23,28,30, 34,36,38,39	37,12,32, 8,25	5				
U8	2F3		2,3,8,14,15,19, 20,22,23,30,34, 36,38,39	37,12,32, 28,25	27				
U9	3F0	3,23	6,8,9,11,12,14, 15,19,20,22,25, 28,32,33,34,36, 38,39	37,30,2	21				

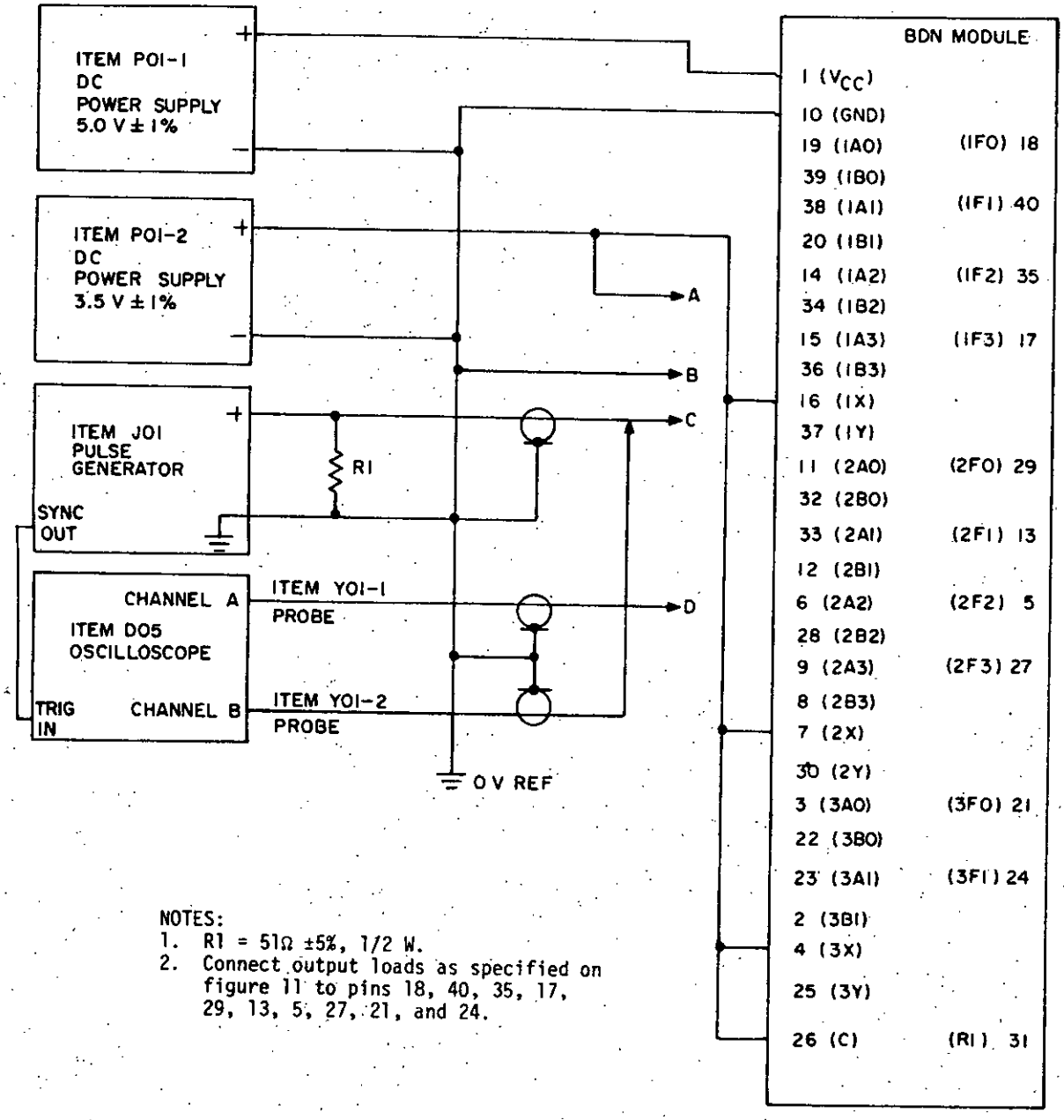
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TABLE XIX. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U10	3F1 Output crosstalk, high level	3,23	2,6,8,9,11,12, 14,15,19,20,25, 28,32,33,34,36, 38,39	37,30,22	24	0	400	mV	20,21

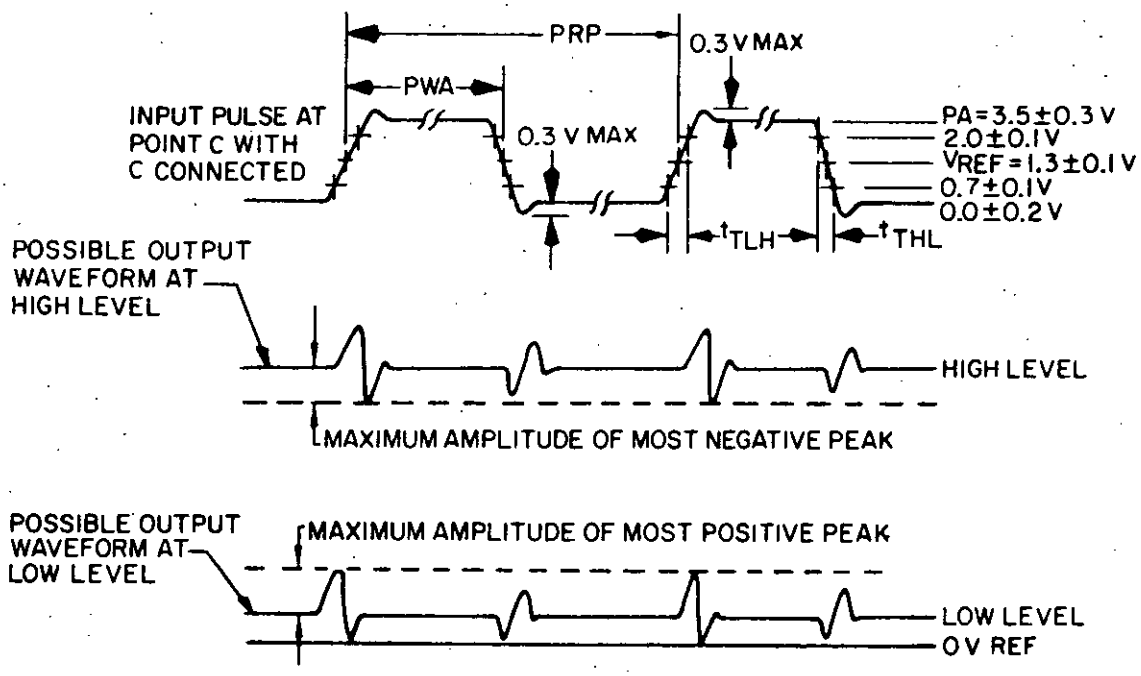
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NOTES:
1. RI = 51Ω ±5%, 1/2 W.
2. Connect output loads as specified on figure 11 to pins 18, 40, 35, 17, 29, 13, 5, 27, 21, and 24.

FIGURE 20. Output crosstalk (low-level and high-level) test circuit for tables XVIII and XIX.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1.0 ±0.1 µs.
2. The lead length of electrical conductors between resistor R1 and input pins; between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pin, shall be less than 1 inch (25.4 mm).

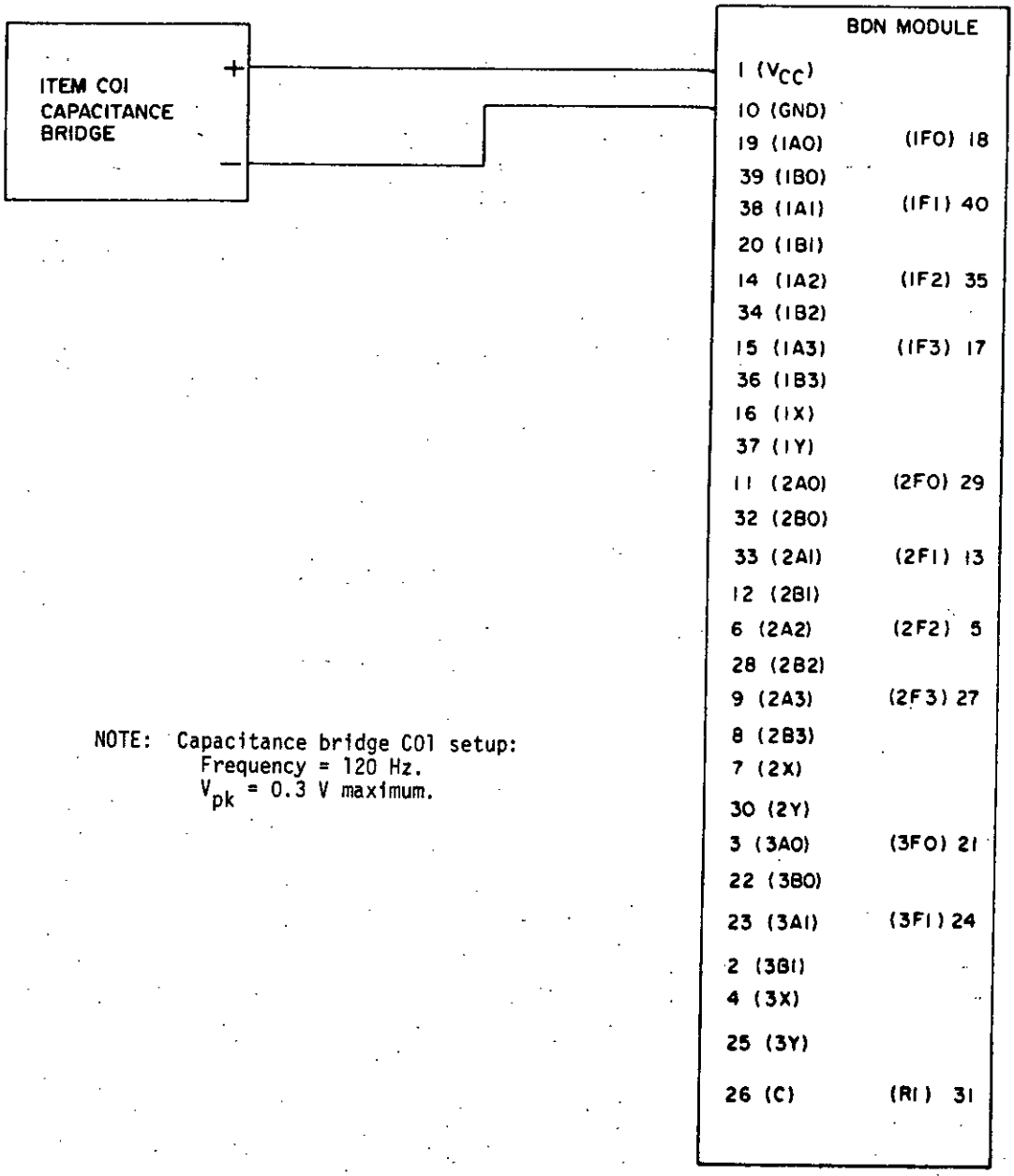
FIGURE 21. Output crosstalk (low-level and high-level) test circuit waveforms for tables XVIII and XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply filter capacitance	1.6	3.0	µF	22

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NOTE: Capacitance bridge C01 setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 22. Power supply filter capacitance test circuit for table XX.

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TABLE XXI. Input capacitance test.

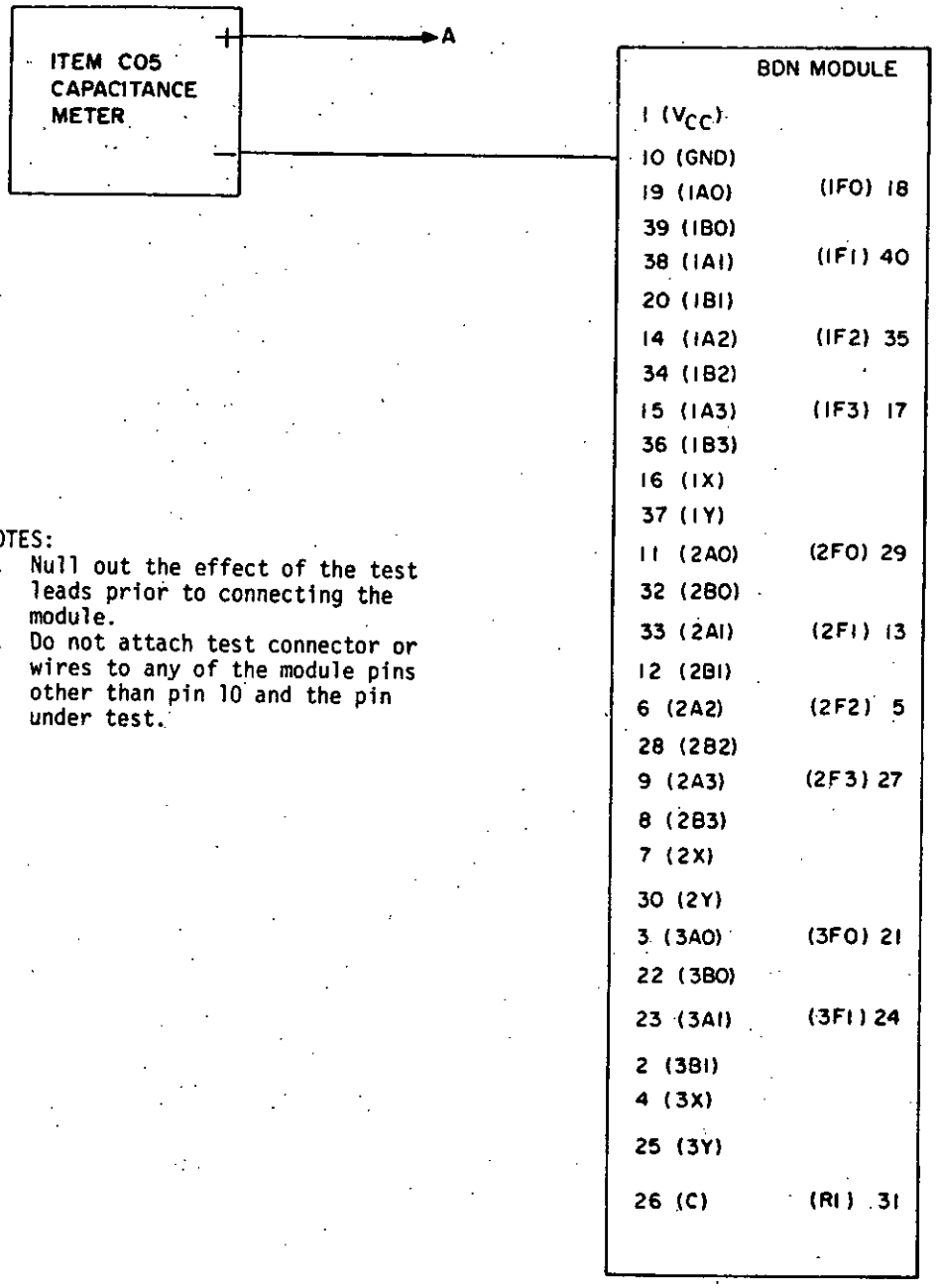
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W1	1A2 Input capacitance	14	2	15	pF	23
W2	1A3	15				
W3	1A0	19				
W4	1A1	38				
W5	2A2	6				
W6	2A3	9				
W7	2A0	11				
W8	2A1	33				
W9	3A0	3				
W10	3X	4				
W11	3A1	23				
W12	3Y	25				
W13	C	26		30		
W14	1B1	20				
W15	1B2	34				
W16	1B3	36				
W17	1B0	39				
W18	1Y	37				
W19	2B3	8				

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TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W20	2B1 Input capacitance	12	2	30	pF	23
W21	2B2	28				
W22	2Y	30				
W23	2BO	32				
W24	3B1	2				
W25	3BO	22				
W26	1X	16				
W27	2X	7				

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test.

FIGURE 23. Input capacitance test circuit for table XXI.

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